Credit Risk & IFRS 9 ECL Engine

Model Validation and Quantitative Assessment Report

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Lending Club Consumer Loans · 2007–2018 · N = 2,260,701 accepted

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Abstract

This report presents the model development, validation, and risk quantification for a retail credit underwriting engine trained on the LendingClub 2007–2018 loan portfolio (2,260,701 accepted loans, of which 1,369,566 have a resolved good/bad outcome and 992,661 form the modelling population after the train/OOT grey zone is excluded; 27,648,741 rejected applications). The Probability of Default (PD) scorecard achieves an out-of-time Gini of **0.4009** (AUC = 0.7004), with Population Stability Index PSI = 0.0031 (model stability: GREEN). The two-stage Loss Given Default model yields mean LGD = 0.8931 and downturn LGD = 1.0000 (90th percentile, Basel-conservative). Basel IRB Risk-Weighted Assets total $6,033,164,733 at 161.8% RWA density. IFRS 9 Expected Credit Loss provisions total $1,209,201,391 (coverage: 32.420%).

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1 Introduction and Executive Summary

Retail credit risk requires modeling frameworks that are explainable and compliant with capital and accounting regulation. Under the Basel Committee on Banking Supervision (BCBS) capital accords (Basel Committee on Banking Supervision 2004) and the International Financial Reporting Standards (IFRS 9) accounting guidelines (International Accounting Standards Board 2014), financial institutions must deploy internal risk engines to assess risk-adjusted pricing, regulatory capital, and forward-looking impairment provisions.

This report documents the mathematical foundation, development methodology, and empirical validation of an end-to-end retail credit underwriting, capital calculation, and expected credit loss (ECL) engine. Trained on 992,661 historical consumer records originated between 2007 and 2018 (drawn from 2,260,701 accepted loans, 1,369,566 of which have a resolved credit outcome), this framework is designed to bridge the gap between risk underwriting, regulatory capital management, and standard accounting provisions.

The core objective is to move away from simplistic statistical estimations and instead construct a highly transparent, mathematically rigorous portfolio risk model that covers:

1. **Underwriting:** An interpretable credit scorecard based on Weight of Evidence (WoE) monotonic binning and regularised logistic regression, validated against a non-linear LightGBM challenger model.
2. **Loss Mitigation:** A bimodal two-stage LGD model (cure probability + conditional severity) with Downturn LGD adjustments for capital stress testing.
3. **Capital Reserve:** Per-loan and aggregated Basel IRB regulatory capital calculation under the "Other Retail" Vasicek Asymptotic Single Risk Factor (ASRF) model.
4. **Financial Provisioning:** A forward-looking IFRS 9 expected credit loss engine driven by discrete-time logistic survival curves across three stages, probability-weighted across macroeconomic scenarios.
5. **Decision Optimisation:** A profit-maximising score cut-off model and reject inference (parcelling) methodology to address selection bias in the underwriting population.

1.1 Key Performance Metrics

Table 1 summarizes the portfolio-level credit metrics and validation statistics calculated across the underwriting, capital, and impairment phases.

An OOT AUC of 0.7004 (Gini 0.4009) is the realistic discrimination ceiling for an application scorecard built on origination-only features on LendingClub-style unsecured consumer data: $0.30 - 0.45$ is the published Gini range for consumer-credit scorecards surveyed by Lessmann et al. (2015), while the tree challengers reach 0.7039 on the same features, indicating the scorecard sits modestly below the attainable ceiling rather than at it. The figure should be read as realistic, benchmarked performance rather than a shortfall.

The scorecard demonstrates stable, high-contrast risk separation across distinct macroeconomic cycles. Under Basel capital standards, the risk-sensitive Internal Ratings-Based (IRB) approach identifies a capital requirement of **$482,653,179** compared to **$223,786,919** under the Standardised Approach. This capital surcharge of **$258,866,260** reflects the elevated risk profile (PD/LGD) of the LendingClub consumer portfolio, demonstrating that a flat 75% risk weight under the Standardised Approach materially undercapitalises this retail asset class.

This risk-sensitive capital surcharge highlights the necessity of developing internal ratings-based (IRB) risk frameworks (Basel Committee on Banking Supervision 2004) to ensure adequate capital provisioning for higher-yielding, higher-risk retail credit assets rather than relying on rigid, risk-insensitive standardized approaches.

TABLE 1. Portfolio Headline Summary and Quantitative Benchmarks

Quant Dimension	Metric Value	Regulatory Purpose & Benchmark
PD Discrimination (OOT)	0.4009 Gini	Out-of-Time score risk rank-ordering capability.
	(0.7004 AUC)	Matches regulatory standards for acceptable discrimination.
OOT Separation (KS)	0.2895	Kolmogorov-Smirnov statistic; values above ~ 0.30 are typical for retail scorecards.
OOT Calibration Brier	0.1796	Brier score reflecting high probability accuracy.
OOT Calibration p-value (raw, pre-recalib.)	0.0000	Hosmer-Lemeshow p-value on the **raw** scorecard PD ($p > 0.05$ confirms good calibration); see Table 17 for the post-recalibration before/after comparison.
OOT Population Stability	0.0031	PSI Train-to-OOT (< 0.10 denotes absolute population stability).
LGD Model Summary	0.8931 Mean LGD	Primary provisioning loss rate.
	1.0000 Downturn	Conservative 90th percentile stress limit for capital charge.
Expected Loss (EL)	$279,690,144	Lifetime Expected Loss projection of current portfolio.
Portfolio EL Rate	7.50%	Underwriting expected loss density.
Basel IRB Total RWA	$6,033,164,733	Risk-Weighted Assets calculated using retail ASRF formula.
Basel SA Total RWA	$2,797,336,487	Standardised Approach baseline capital RWA reference (75% RW).
Basel RWA Density	161.8%	IRB RWA divided by total portfolio EAD ($3,729,781,983).
Portfolio IFRS 9 ECL	$1,209,201,391	Probability-weighted Stage 1, 2, & 3 Expected Credit Loss.
Portfolio ECL Coverage	32.420%	Capital coverage buffer (Total ECL / Total EAD).

2 DATA ENGINEERING AND EXPLORATORY ANALYSIS

2.1 DATA SOURCE AND EXCLUSIONS

The primary underwriting and historical performance data is derived from Lending Club's consumer loan database, covering the years 2007 through 2018. This dataset contains credit bureau features and demographic information gathered at loan origination, along with post-origination transaction and default markers.

To ensure methodological correctness, loans with ambiguous or immature repayment statuses are excluded from the modeling population. Specifically, loans marked as *"Current"*, *"In Grace Period"*, or *"Late (16–30 days)"* are removed since their ultimate credit outcome is unresolved. The remaining loans represent the underwriting and model development population.

2.2 TARGET DEFINITION (PD)

A binary default indicator (Y) is defined on each loan's *terminal resolved status* at the 2018Q4 snapshot, taken directly from the pipeline configuration:

$$Y = \begin{cases} 1 \text{ (Bad)}, & \text{if status } \in \{\text{Charged Off, Default, Does not meet the credit policy. Status:Charged Off, Late (3} \\ 0 \text{ (Good)}, & \text{if status } \in \{\text{Fully Paid, Does not meet the credit policy. Status:Fully Paid}\} \end{cases} \quad (1)$$

The bad set includes *Late (31–120 days)* and is therefore **wider** than the BCBS 90+ DPD reference definition: it admits 31–89 DPD delinquency as default. This is deliberate. The dataset is a status snapshot rather than a days-past-due panel, so loans sitting at exactly 90+ DPD on the snapshot date are few, and restricting the bad set to them would discard most of the delinquent population. The consequence is a *more conservative* default flag than the regulatory standard, and it should be read that way wherever this report compares its default rates to published benchmarks (Section 10).

2.3 Out-of-Time (OOT) Splitting

To replicate standard banking validation practices, the data is split chronologically based on loan origination date (`issue_d`):

- **Training Population:** Loans originated prior to January 2015 (N = 363,317; 36.60% of modelling population). Training bad rate: 17.09%.
- **In-Time Holdout Set:** Simple random 20% sample from the training period (N = 90,829; 9.15% of modelling population). The sample is drawn without class stratification; at this size the 82.91%/17.09% good/bad ratio is reproduced by the law of large numbers rather than enforced by design.
- **Out-of-Time (OOT) Set:** All loans originated between January 2016 and December 2018 (N = 538,515; 54.25% of modelling population). OOT observed bad rate: 26.33%, reflecting the portfolio's seasoning and the macroeconomic deterioration of the 2016–2018 vintage cohorts.

This chronological split simulates how the model will perform on future vintages, testing for structural or macroeconomic shifts.

Excluded grey zone. The two cut-offs are deliberately non-adjacent: loans originated in the twelve months between them (the entire **2015** origination cohort) fall into neither partition and are *dropped from the modelling population entirely*. The embargo prevents 2015 vintages — which are simultaneously close enough to the training window to share its underwriting regime and immature enough at the 2018Q4 snapshot to have unresolved outcomes — from contaminating either side of the temporal validation. The percentages above, the modelling population of 992,661 loans (376,905 resolved-outcome loans fall in the grey zone and are dropped), and every vintage table in this report are therefore stated *net of 2015*, which is why the vintage series in Tables 13 and 18 step directly from 2014 to 2016.

2.4 Leakage and Target Variable Separation

Data leakage is controlled by dropping post-origination fields (e.g., outstanding balance, payment records, and recovery metrics) from the PD feature set. However, these post-origination variables (such as actual recoveries and write-offs) are preserved for the defaulted-only population. This allows the LGD model to evaluate recovery performance without leaking future data into the PD scorecard.

2.5 Vintage Cohort Analysis

Vintage analysis tracks the cumulative default curves of origination cohorts (quarters) over their Months-on-Book (MOB). Figure 1 illustrates these curves. A steep initial slope indicates seasoning, while the curves flatten as high-risk accounts default early, illustrating standard credit risk dynamics.

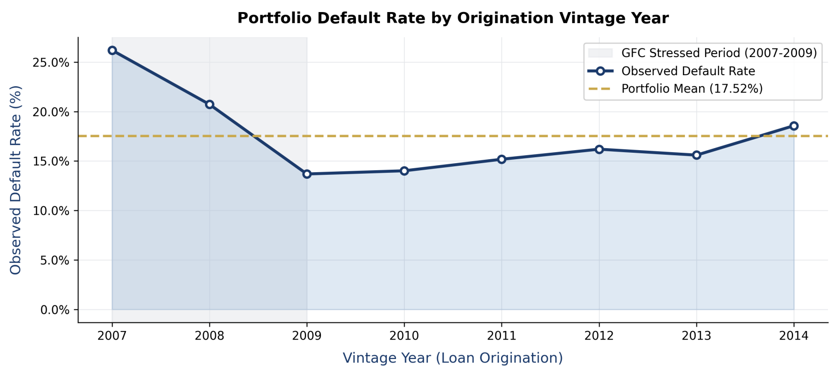
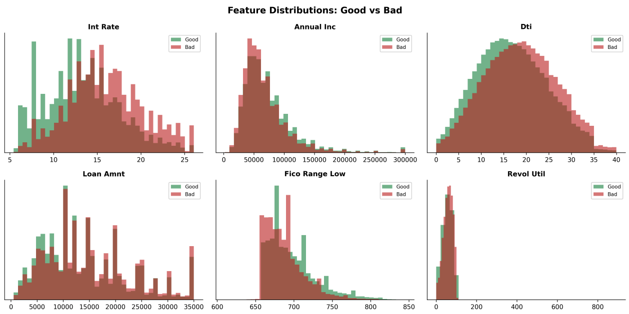
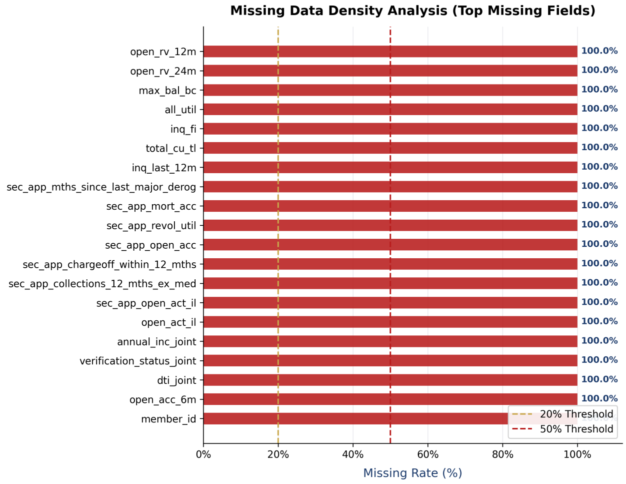


FIGURE 1. Cumulative Default Rates by Quarterly Vintage Cohorts

Figure 3 displays additional exploratory distributions, demonstrating the relationship between historical default rates and risk grades, amortization terms, and loan purposes. Figure 2 completes the exploratory picture with the marginal distributions of the key numeric predictors and the missingness profile of the raw feature set; the treatment applied to those missing values is described in Section 3.1.



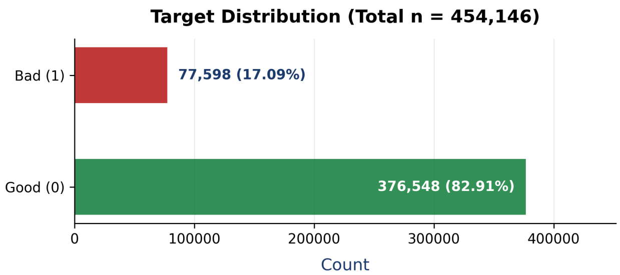
(a) Key numeric feature distributions



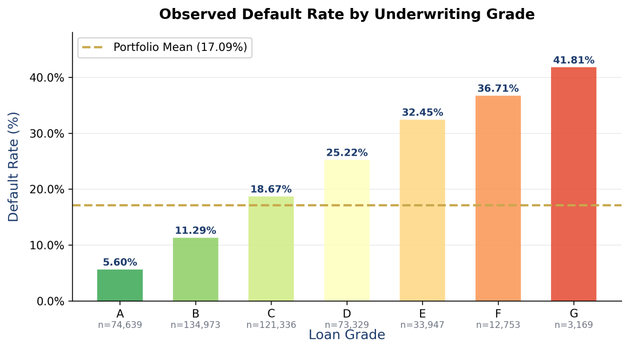
(b) Missingness density

Figure 2. Exploratory data analysis of the development sample: predictor distributions and missingness.

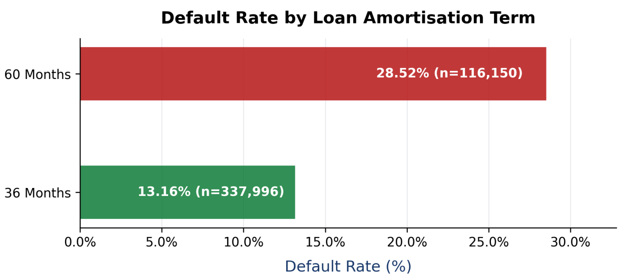
All EDA visualizations in this section are computed on the in-time development sample (train and test partitions of the resolved-outcome modelling population), not the full raw portfolio: the Out-of-Time partition is withheld from all exploratory analysis to preserve the integrity of the temporal validation, and loans with unresolved statuses are excluded per Section 2.1. Portfolio-level capital and impairment calculations are run on the modelling population ($N = 992{,}661$), i.e. the train, test and OOT partitions combined — not on the 2,260,701 loans in the source file, since loans with an unresolved status carry no good/bad label and are excluded (1,369,566 remain after that filter, of which 376,905 fall in the excluded grey zone described in Section 2.3). Population counts embedded in the EDA figures therefore reflect the development sample.



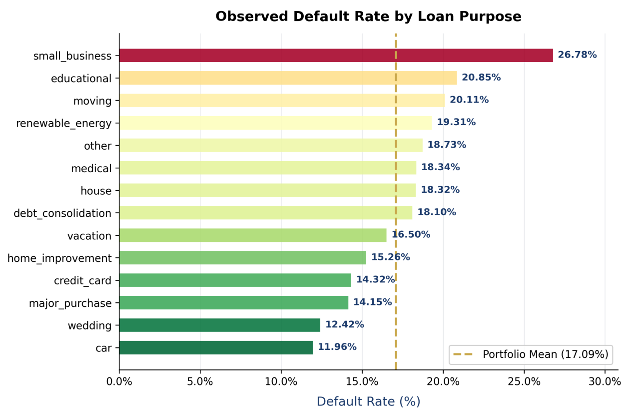
(a) Target Distribution (Good vs. Bad)



(b) Default Rate by Underwriting Grade



(c) Default Rate by Amortisation Term



(d) Default Rate by Loan Purpose

Figure 3. EDA Risk, Grade, Term, and Purpose Distributions

3 Probability of Default (PD) Scorecard Development

3.1 Weight of Evidence (WoE) and Information Value (IV)

Continuous features are binned to handle non-linear relationships, outliers, and missing values. Binning is performed by `optbinning`'s optimal-binning solver, which searches bin boundaries under a mixed-integer formulation; where the solver is unavailable the pipeline falls back to a quantile-split binner that merges adjacent bins until the sign of the WoE differences is homogeneous. Which of the two produced the bins reported here is recorded in Appendix C. For each bin i, the Weight of Evidence (WoE) is calculated as:

$$WoE_i = \ln\left(\frac{\text{Proportion of Good}_i}{\text{Proportion of Bad}_i}\right) = \ln\left(\frac{N_{G,i}/N_{G,total}}{N_{B,i}/N_{B,total}}\right) \tag{2}$$

The predictive power of each feature is evaluated using the Information Value (IV):

$$IV = \sum_{i=1}^{k}\left(\frac{N_{G,i}}{N_{G,total}} - \frac{N_{B,i}}{N_{B,total}}\right) \times WoE_i \tag{3}$$

(Siddiqi 2017, Ch. 4); (Hand and Henley 1997). Features with an IV below 0.02 are dropped due to low predictive power, while multicollinearity is controlled by removing features with a Variance Inflation Factor (VIF) greater than 5.0. Two further selection stages follow; the complete funnel is set out in Section 3.3.

Missing-value treatment

Missing values are passed to the binner as missing and receive a bin of their own, with a Weight of Evidence estimated from the observed default rate among the missing rows. Their contribution to a borrower's score is therefore whatever the data says it is, and it is visible as a populated row in the points table rather than folded invisibly into another band.

This matters more than it may appear. An earlier version of this pipeline substituted the sentinel value −9999 before binning. That placed every missing observation in the extreme lower bin of its feature, with three consequences: the binner's own "Missing" bin was structurally empty and its code path never executed; the implied risk direction was arbitrary and inconsistent across features, since a missing value scored as *worst* for a feature where low is risky (FICO, available credit) and as *best* for one where low is safe (DTI, recent enquiries); and for `mths_since_recent_bc`, where missing means the borrower has never held a bankcard, the assignment carried the wrong sign outright. Because the two binning implementations also handled the sentinel differently, they produced materially different models from identical data.

3.2 Logistic Regression and Scorecard Scaling

A regularized logistic regression (Hosmer, Lemeshow, and Sturdivant 2013) is fitted on the WoE-transformed features. Since higher WoE corresponds to a higher proportion of "Good" loans relative to "Bad" loans, all coefficients must be negative when predicting default ($Y = 1$). The scorecard is then scaled to a points-based system using:

$$Score = Offset + Factor \times \ln(\text{odds}) \tag{4}$$

$$Factor = \frac{PDO}{\ln(2)}, \quad Offset = TargetScore - Factor \times \ln(TargetOdds) \tag{5}$$

For $TargetScore = 600$ points, $TargetOdds = 50 : 1$, and $PDO = 20$ (points to double the odds), the scaling parameters are:

- $Factor = 28.8539$
- $Offset = 487.123$

The points contributed by a specific binned attribute j are calculated as:

$$Points_j = \left(-(WoE_j \times \beta_j) + \frac{\alpha}{n}\right) \times Factor + \frac{Offset}{n} \tag{6}$$

where β_j is the regression coefficient, α is the model intercept, and n is the number of active features (Anderson 2007, Ch. 5).

3.3 Feature Selection Results: the Four-Stage Funnel

After WoE binning, candidate features pass through **four** sequential filters, not two. Reporting only the first two would leave the final list unreconcilable with the IV ranking below — several high-IV candidates are removed by the later stages:

1. **IV band.** Features with $IV < 0.02$ (negligible predictive power) or $IV > 0.50$ (likely target leakage) are excluded.
2. **VIF filter.** Multicollinearity is controlled by iteratively removing features with a Variance Inflation Factor above 5.0.
3. **ElasticNet shrinkage.** A cross-validated penalised logistic regression (`LogisticRegressionCV`, SAGA solver, ℓ_1 ratio 0.5, 3-fold stratified CV) is fitted on the survivors, and features whose coefficient shrinks to $|\beta| \leq 10^{-4}$ are dropped. This stage removes predictors that are individually informative but redundant given the rest of the set — which is why a high-IV variable can be absent from the final list.
4. **Sign check.** The unpenalised logistic regression is fitted and any feature whose coefficient carries the wrong sign (positive on a WoE predictor, i.e. higher WoE implying higher default risk) is dropped, and the model refitted on the survivors.

The funnel for this run: 83 candidate features → 37 after the IV band $[0.02, 0.50]$ → 25 after the VIF filter → 22 after ElasticNet shrinkage → **17** after the sign check. Dropped by the sign check: `bc_util`, `mo_sin_rcnt_rev_tl_op`, `revol_util_x_open_acc`, `tot_cur_bal`, `total_rev_hi_lim`.

The final set is: `grade_enc`, `term_enc`, `loan_to_income`, `fico_range_high`, `revol_util_x_new_acc`, `new_accounts_ratio`, `dti`, `bc_open_to_buy`, `num_tl_op_past_12m`, `log_annual_inc`, `avg_cur_bal`, `mo_sin_rcnt_tl`, `mths_since_recent_inq`, `inq_last_6mths_per_month`, `num_rev_tl_bal_gt_0`, `mths_since_recent_bc`, `mort_acc`.

Table 2. Feature Information Value (IV) Ranking (top 15)

Feature	IV	Predictive Power	Outcome
`int_rate`	0.4072	Strong	Dropped (VIF)
`grade_enc`	0.3963	Strong	**Retained**
`term`	0.1998	Medium	Dropped (VIF)
`term_enc`	0.1998	Medium	**Retained**
`loan_to_income`	0.1224	Medium	**Retained**
`fico_range_low`	0.1143	Medium	Dropped (VIF)
`fico_range_high`	0.1143	Medium	**Retained**
`revol_util_x_new_acc`	0.0969	Weak	**Retained**
`total_debt_to_income`	0.0650	Weak	Dropped (ElasticNet)
`new_accounts_ratio`	0.0648	Weak	**Retained**
`dti`	0.0555	Weak	**Retained**
`acc_open_past_24mths`	0.0543	Weak	Dropped (VIF)
`bc_open_to_buy`	0.0519	Weak	**Retained**
`num_tl_op_past_12m`	0.0403	Weak	**Retained**
`annual_inc`	0.0365	Weak	Dropped (VIF)

Note: the remaining 22 candidate features all carry IV ≤ 0.0365 and are omitted; the full ranking is in `outputs/scorecard_tables.json`. The *Outcome* column names the stage of the four-stage funnel (Section 3.3) at which each feature left, which is why a high-IV feature can be absent from the final model. Monotone transforms of the same underlying variable (for example a raw and a log-scaled version) carry near-identical IV by construction and are not independent evidence.

3.4 Logistic Regression Coefficient Output

The following table presents the logistic regression coefficient estimates, standard errors, z-statistics and p-values for all retained features. As expected for a WoE-based model predicting default ($Y = 1$), all feature coefficients carry a **negative sign**: higher WoE values indicate a higher proportion of "Good" loans, and therefore map to lower default probability.

Table 3. Logistic Regression Coefficient Summary

Feature	Coefficient	Std. Error	z-stat	p-value
`const`	-1.6110	0.0053	-303.068	0.0000***
`grade_enc`	-0.5155	0.0106	-48.619	0.0000***
`term_enc`	-0.6521	0.0125	-51.996	0.0000***
`loan_to_income`	-0.4488	0.0158	-28.414	0.0000***
`fico_range_high`	-0.2674	0.0190	-14.090	0.0000***
`revol_util_x_new_acc`	-0.0487	0.0240	-2.025	0.0429**
`new_accounts_ratio`	-0.4537	0.0261	-17.415	0.0000***
`dti`	-0.3300	0.0221	-14.915	0.0000***
`bc_open_to_buy`	-0.3963	0.0272	-14.582	0.0000***
`num_tl_op_past_12m`	-0.2284	0.0355	-6.425	0.0000***
`log_annual_inc`	-0.4577	0.0325	-14.074	0.0000***
`avg_cur_bal`	-0.3561	0.0334	-10.669	0.0000***
`mo_sin_rcnt_tl`	0.0182	0.0396	0.460	0.6452
`mths_since_recent_inq`	-0.0281	0.0518	-0.542	0.5877
`inq_last_6mths_per_month`	-0.5141	0.0400	-12.844	0.0000***
`num_rev_tl_bal_gt_0`	-0.3301	0.0347	-9.516	0.0000***
`mths_since_recent_bc`	-0.3773	0.0378	-9.992	0.0000***
`mort_acc`	-0.3900	0.0424	-9.201	0.0000***

Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$

3.5 Scorecard Points Table

Each WoE bin is converted to credit score points using the scaling formula. Higher points correspond to lower default risk. The table below shows the top bins ranked by point spread (i.e., the features with the greatest discriminatory contribution to the final score).

Table 4. Credit Scorecard Points Table — complete bin ladders for the highest-spread features

Feature	Bin	WoE	β	N	DR%	Points
`grade_enc`	(-inf, 1.50)	1.2418	-0.5155	59,675	5.62%	44.4
	[1.50, 2.50)	0.4866	-0.5155	107,931	11.25%	33.2
	[2.50, 3.50)	-0.1063	-0.5155	96,905	18.65%	24.3
	[3.50, 4.50)	-0.4971	-0.5155	58,848	25.31%	18.5
	[4.50, inf)	-0.9212	-0.5155	39,958	34.12%	12.2
	Special	0.0000	-0.5155	0	0.00%	25.9
	Missing	0.0000	-0.5155	0	0.00%	25.9
`term_enc`	(-inf, 48.00)	0.3076	-0.6521	270,363	13.16%	31.7
	[48.00, inf)	-0.6604	-0.6521	92,954	28.52%	13.5
	Special	0.0000	-0.6521	0	0.00%	25.9
	Missing	0.0000	-0.6521	0	0.00%	25.9

WoE = ln(%Good/%Bad); Points = $(-\mathrm{WoE}_j * \beta_j + \alpha/n)$ * Factor + Offset/n. The remaining 15 scorecard features follow the same construction; their complete ladders are in `outputs/scorecard_tables.json`. Where a calibrator is attached in production, the score-to-PD mapping tabulated here is the *pre-recalibration* relationship: the points scale is built from the raw logit so that it stays additive and auditable, while the deployed PD passes through the recalibration transform of Section 7.2.

3.6 Interpretability vs. Performance: Challenger Model Benchmark

A non-linear LightGBM model was trained as a challenger (Baesens, Roesch, and Scheule 2016) on the scorecard's selected predictors (see Section 7.8 for the exact feature count actually consumed by each model). Although gradient boosting is competitive in-sample, on the out-of-time set the best challenger, Weighted Ensemble, *exceeds* the scorecard's discrimination (OOT AUC 0.7039 vs 0.7004), a gap tested in Section 7.8. The scorecard is nonetheless retained as the production model for regulatory and explainability reasons set out below, with the discrimination sacrifice reported rather than obscured. Regulatory standards (such as the US Fair Credit Reporting Act) require financial institutions to provide clear "adverse action codes" (reasons for denial) to rejected applicants. A linear scorecard allows for immediate, exact points-attribution for each feature, which is not possible with complex machine learning models without relying on approximations like SHAP.

To address potential policy-decision circularity from using LendingClub's own underwriting variables (`int_rate` and `grade`), we developed a secondary Pure Underwriting Scorecard (Model B). Model B excludes `int_rate`, `grade_enc`, `grade`, `sub_grade`, `sub_grade_enc`, `loan_amnt`, `funded_amnt`, `funded_amnt_inv`, `installment`. The exclusion list is therefore wider than pricing alone: the requested loan amount and its derived instalment are also withheld, so Model B is a deliberately conservative lower bound on what an independent underwriter could achieve, not a like-for-like re-fit with pricing removed. One qualification belongs here: the engineered `loan_to_income` ratio is *not* excluded, so loan size still reaches Model B in normalised form. Table 5 compares the performance of the full scorecard (Model A) against this independent underwriting model. While Model A is the designated pipeline champion due to its superior discrimination, it utilizes LendingClub's pricing variables which are themselves highly correlated risk assessments. This introduces a degree of decision circularity. Model B (Pure Underwriting Scorecard) shows that a model built solely on raw credit bureau and demographic features remains competitive (OOT AUC = 0.6851 vs Model A's 0.7004, both as reported in Table 5), supporting the scorecard's viability in an independent bank underwriting environment.

Table 5. Champion (Model A) vs. Pure Underwriting Challenger (Model B) Performance

Model	Features Included	OOT AUC	OOT Gini
Model A (Full Scorecard)	Bureau + Application + int_rate + grade	0.7004	0.4009
Model B (Underwriting)	Bureau + Application (excludes int_rate/grade)	0.6851	0.3701

3.7 Selection Bias and Reject Inference (Parcelling)

Scorecard models developed only on approved applicants suffer from selection bias. Because rejected applicants are excluded, their risk profiles and actual default rates are unobserved. To adjust for this, we implemented the **Parcelling** reject inference technique to probabilistically allocate outcomes to rejected applicants based on the accepts scorecard's predictions.

The pooled accepts and parcelled rejects population was refitted to produce a corrected through-the-door (TTD) scorecard. The comparison is restricted to the 2 of 17 scorecard predictors that are genuinely observed for rejected applicants, and both Gini figures are computed on that same set with the same weighting. The remaining 15 exist only in the accepted file and were previously mean-imputed to a training constant across the entire reject population. A constant predictor has no discriminatory power, so the earlier Gini drop measured that imputation at least as much as any latent risk in the through-the-door population, and should not be read as economic evidence about rejected applicants. Table 6 outlines the results of the refitting and selection bias adjustment:

The Gini shift of **-0.0665** points is consistent with the standard credit-cycle finding that through-the-door populations carry higher latent risk. *Measurement caveat:* the shift is computed *within* the parcelled refit — the same through-the-door model scored on the accepts subset versus on the full accepts-plus-parcelled population — and both figures are in-sample; the rejects additionally carry inferred, not observed, labels. The shift is then applied to the champion scorecard's held-out Gini to obtain the through-the-door figure in the table. The two rows are therefore not two independently validated scorecards, and the shift should be read as a directional selection-bias indicator rather than a measured out-of-sample degradation.

Table 6. Reject Inference (Parcelling) Gini Coefficient Shift

Scorecard Population	Gini	Business Interpretation
Accepts-Only (Base)	0.4039	Champion scorecard on the approved-only in-time test partition.
Through-the-Door (Parcelled Refit)	0.3375	Corrected scorecard accounts for selection bias across TTD.
Gini Coefficient Shift	-0.0665	Conservative risk dilution when scoring raw through-the-door applicants.

3.8 Survival Analysis: Kaplan-Meier and Cox Proportional Hazards

The production PD term structure (Section 6) is a discrete-time hazard model. As a challenger model we additionally fit a time-to-event survival model, the industry standard for IFRS 9 lifetime-PD term-structure work (Bellotti and J. Crook 2009). Kaplan-Meier estimators give non-parametric survival curves $S(t)$ per credit grade (Figure 4), and a Cox proportional-hazards model quantifies each covariate's multiplicative effect on the default hazard. The duration is a months-on-book proxy derived from cumulative payments and the event is the binary default flag, a synthesised time-to-event dataset since the raw data records no observed default month (a documented limitation revisited in Section 10). The model's rank-discrimination is summarised by the concordance index (Cox C-index = 0.6439), the survival-analysis analogue of the AUC. The production hazard model attains an OOT AUC of 0.6847 (Gini 0.3693); for a binary outcome this equals its concordance index, so it is directly comparable to the Cox challenger's C-index of 0.6439. Reporting it closes a gap in which the champion — the model that produces every lifetime PD in the ECL — was the only one carrying no measure of rank-ordering ability, while its challenger did.

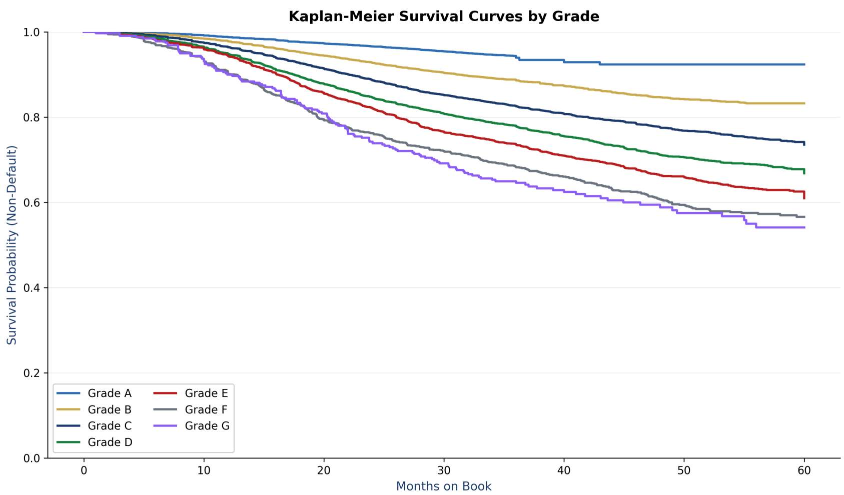


Figure 4. Kaplan-Meier non-default survival curves by credit grade. Lower grades separate downward, confirming the expected monotone grade–risk ordering.

Table 7 reports the fitted Cox coefficients, hazard ratios $\exp(\beta)$ and Wald p-values. A hazard ratio above 1 raises the instantaneous default hazard; below 1 lowers it. The dominant hazard multipliers attach to credit grade and interest rate, while debt-to-income and amortisation term enter with smaller but individually significant positive coefficients (all values as tabulated below); each covariate therefore raises the default hazard monotonically as it increases, consistent with the scorecard's risk ordering.

TABLE 7. Cox Proportional-Hazards Model — Covariate Summary

Covariate	Coef (β)	Hazard Ratio	p-value
Credit Grade (A=1..G=7)	+0.12773	1.13625	<0.001
Interest Rate	+0.03797	1.03870	<0.001
Debt-to-Income (DTI)	+0.01317	1.01325	<0.001
Term (months)	+0.00314	1.00314	<0.001

4 LOSS GIVEN DEFAULT (LGD) AND EXPOSURE AT DEFAULT (EAD)

4.1 BIMODAL TWO-STAGE LGD MODEL

Loss Given Default (LGD) represents the economic loss rate incurred when an exposure defaults. Unlike PD, which models a binary outcome, LGD is continuous, bounded within $[0, 1]$, and heavily bimodal. Defaults typically result in either a complete recovery (LGD = 0, "cure") or a near-total loss (LGD close to 1).

To capture this bimodal behavior (Schuermann 2004), a **two-stage LGD model** is constructed as one of two candidate severity models — benchmarked against a LightGBM challenger in Section 4.2, with the lower out-of-sample error determining which model is deployed:

1. **Stage 1 (Cure Model):** A gradient-boosted classifier (100 trees, depth 4, learning rate 0.05, class-balanced sample weights) estimates the probability of a zero-loss outcome (cure). Writing $g(\mathbf{x})$ for its fitted cure probability, the loss probability is:

$$p_{\text{loss}} = P(\text{LGD} > 0 \mid \text{Default}) = 1 - g(\mathbf{x}) \tag{7}$$

   A tree ensemble is used rather than a logistic link because the cure indicator responds non-monotonically to recovery-related covariates; the severity stage below remains a GLM, so the deployed LGD is a hybrid rather than a pure two-stage regression.

2. **Stage 2 (Severity Model):** For defaulted loans that incur a loss (LGD $>$ 0), a fractional logit GLM (Binomial family, logit link) models the conditional loss severity (Papke and Wooldridge 1996; Bellotti and J. Crook 2012):

$$E[\text{LGD}|\text{LGD} > 0] = \frac{1}{1 + e^{-\mathbf{x}'\beta}} \tag{8}$$

The final predicted LGD is the product of the two stages:

$$\text{LGD}_{\text{pred}} = p_{\text{loss}} \times E[\text{LGD}|\text{LGD} > 0] \tag{9}$$

For Basel IRB capital calculations, a **Downturn LGD** is taken as the 90th percentile of the *realised* severity distribution across loss-incurring defaults (floored at the mean predicted LGD, so the stress can never fall below the central estimate):

- **Mean Expected LGD (deployed model):** 0.8931 (used for IFRS 9 Stage 1 & 2 provisions)
- **Downturn LGD:** 1.0000 (used for Basel RWA capital calculations)

On the magnitude of the downturn uplift. On this fully unsecured, non-revolving instalment book the realised severity of loss-incurring defaults is concentrated at total loss, so the 90th percentile sits at the cap of the $[0, 1]$ range (1.0000), 10.69 pp above mean LGD. Two consequences follow. It is a real empirical percentile rather than a modelled stress — more than a tenth of loss-incurring defaults recover nothing at all. And because severity cannot exceed total loss, the Basel capital calculation has no headroom above this figure: any further conservatism must come from PD or EAD, not LGD.

Why Mean LGD Is Close to Total Loss

The realised LGD used to fit and validate the severity model is computed directly from resolved loan outcomes as

$$\text{LGD}_{\text{realised}} = 1 - \frac{\max(0,\ \text{recoveries} - \text{collection\_recovery\_fee})}{\max(1,\ \text{funded\_amnt} - \text{total\_rec\_prncp})} \tag{10}$$

i.e. net recoveries (gross post-charge-off recoveries less the fee paid to the collection agency) divided by the EAD proxy (funded amount less principal already repaid at charge-off), clipped to $[0, 1]$. On this fully unsecured, non-revolving instalment book, post-charge-off cash recoveries are small relative to the outstanding balance, so realised severity concentrates near total loss; the two-stage model's cure probability p_{loss} separately absorbs the loans that resolve with zero loss, while the conditional severity stage captures how close to total the loss is for the remainder. The published unsecured LGD bands cited below (Schuermann (2004): 0.45–0.55, corporate/wholesale debt; Bellotti and J. Crook (2012): 0.25–0.45, revolving retail cards) are measured on portfolios with active collections/settlement programmes and revolving-card structures where partial recoveries are more common; the deployed mean LGD of 0.8931, measured on already-charged-off, non-revolving loans against a book-value EAD proxy with no collateral, is a structurally different (and higher) quantity by construction rather than an anomaly. Its magnitude relative to these bands is discussed further, with numeric verdicts, in the benchmark table (Section 10.5).

LGD Benchmark Context: Two severity models are compared on a chronologically held-out selection sample — a two-stage cure-plus-severity model and a LightGBM challenger — and the model with the lower error there is deployed for all downstream provisioning and capital; the metrics in Table 8 are then computed on a disjoint reporting sample so the published out-of-sample performance is not measured on the same defaults used for the promotion decision. For unsecured LGD, Schuermann (2004) report a 0.45–0.55 range (a corporate/wholesale-debt review) while Bellotti and J. Crook (2012) document a lower 0.25–0.45 band for revolving retail credit-card portfolios. The deployed mean LGD of 0.8931 and downturn LGD of 1.0000 are assessed against these published ranges in the benchmark table (Section 10.5), where any deviation is reported together with its driver. The Downturn LGD is taken at the 90th percentile of the realised severity distribution and used exclusively for Basel IRB capital calculations, providing a buffer of 10.69 pp above the mean.

4.2 Out-of-Sample LGD Validation

The severity model is validated out-of-time on defaulted loans from vintages held out of the fitting window. Table 8 reports the standard LGD backtesting metrics — Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), the coefficient of determination R^2, and a two-sample Kolmogorov-Smirnov (KS) statistic comparing the marginal predicted and realised LGD distributions (Loterman et al. 2012). The KS statistic is reported without its p-value: at this sample size ($n = 74{,}981$) the test is hyper-sensitive to any trivial distributional difference and, being a marginal (not per-loan) comparison, cannot by itself certify calibration; Figure 5 and the decile table against the 45° line are the calibration evidence. The aggregate portfolio-level mean LGD sits above the unsecured LGD literature ranges cited in Section 10.5 (driver discussed there), and the loan-level predictive performance of the deployed severity model is separately weak, with a negative out-of-sample R^2 of -0.0709 (Table 8); the rejected two-stage model was materially worse at $R^2 \approx -2.70$, which drove the champion–challenger switch. A negative R^2 is a well-documented model risk: realized retail LGD is highly bimodal (concentrated at 0 for cured loans and near 1.0 for write-offs), and predicting the exact loss severity on unsecured, non-collateralized consumer loans is statistically challenging. While the model remains suitable for calculating conservative portfolio-level capital buffers, its loan-level predictions should be treated with caution.

TABLE 8. Out-of-Sample LGD Validation Metrics ($n = 74, 981$ held-out defaults)

Metric	Value
Mean Absolute Error (MAE)	0.0763
Root Mean Squared Error (RMSE)	0.1080
Coefficient of Determination (R^2)	-0.0709
KS Statistic (marginal dist., pred vs actual)	0.5490

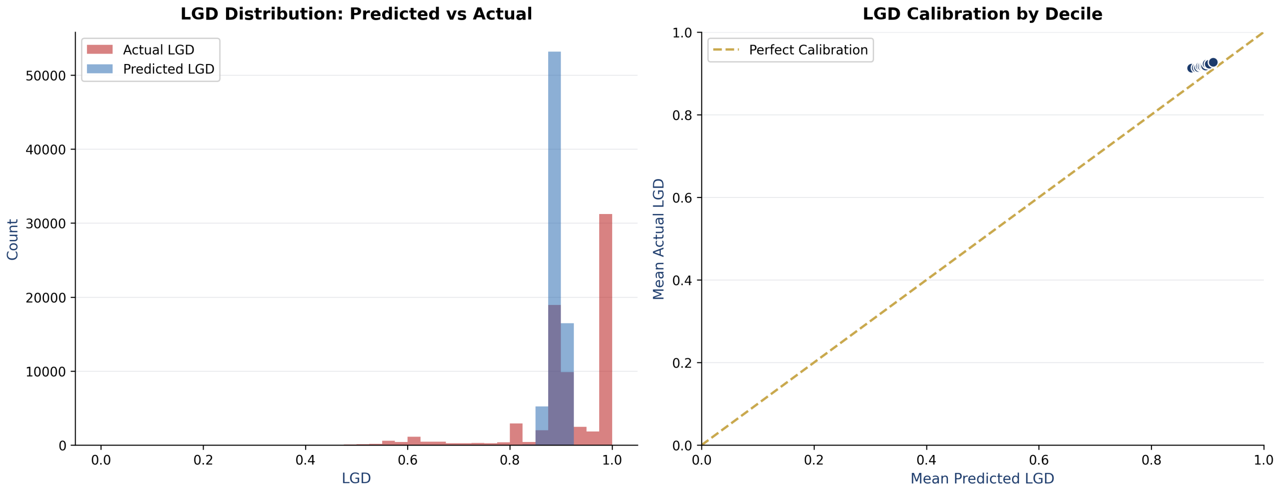


Figure 5. LGD validation: (left) predicted vs realised LGD distributions; (right) mean realised vs mean predicted LGD by predicted decile against perfect calibration.

4.3 Amortisation-Based EAD for Term Loans

Exposure at Default (EAD) represents the outstanding gross balance owed by the borrower at the moment of default. For fully-drawn, non-revolving consumer installment loans, the outstanding principal amortizes deterministically over time. EAD is modeled using a closed-form annuity amortization formula:

$$EAD(t) = \text{funded\_amnt} \times \frac{1-(1+r)^{-(T-t)}}{1-(1+r)^{-T}} \tag{11}$$

where r is the monthly interest rate on the loan contract, T is the original term in months, and t is the elapsed Months-on-Book (MOB) at default.

For revolving credit facilities (such as credit cards or overdraft limits), EAD must capture future drawdowns using a Credit Conversion Factor (CCF):

$$EAD = \text{Drawn Balance} + \text{CCF} \times \text{Undrawn Limit} \tag{12}$$

Since Lending Club loans are fully drawn installment loans with no revolving limits, the undrawn limit is zero. Using the closed-form amortization formula to calculate outstanding principal is an appropriate simplification, which is fully documented and standard in retail banking.

5 Basel IRB Capital & Capital Stress Testing

Throughout this report, $Z < 0$ corresponds to an adverse macroeconomic shock (recession); $Z > 0$ corresponds to a favourable shock (expansion). This sign convention applies uniformly to the Vasicek stress test below, the IFRS 9 macro-scenario mapping (Table 14), and the ECL sensitivity analysis (Figure 12a).

5.1 ASRF Vasicek Model and the "Other Retail" Formula

Under the Basel II/III framework (Basel Committee on Banking Supervision 2004), banks calculate capital requirements for retail exposures using the Asymptotic Single Risk Factor (ASRF) model (Vasicek 2002). This framework assumes that portfolio risk is driven by a single systematic macroeconomic factor. The retail supervisory correlation (R) and capital requirement (K) formulas for the "Other Retail" asset class are:

$$R = 0.03 \times \frac{1-e^{-35\times PD}}{1-e^{-35}} + 0.16 \times \left[1 - \frac{1-e^{-35\times PD}}{1-e^{-35}}\right] \tag{13}$$

$$K = \text{Downturn LGD} \times \Phi\left(\frac{\Phi^{-1}(PD) + \sqrt{R}\,\Phi^{-1}(0.999)}{\sqrt{1-R}}\right) - PD \times \text{Downturn LGD} \tag{14}$$

(Basel Committee on Banking Supervision 2004, §328). Where Φ is the standard normal cumulative distribution function, Φ^{-1} is the inverse standard normal CDF, and the confidence level is set to **99.9%**. Risk-Weighted Assets (RWA) are calculated by scaling the capital requirement (K):

$$RWA = K \times 12.5 \times EAD \tag{15}$$

PD horizon. The PD entering this formula is a **one-year** probability of default, as §328 requires. The scorecard's own target is the loan's terminal resolved status, so its direct output is a *lifetime* PD; it is converted at the point of use under a constant marginal-hazard assumption over the remaining term,

$$PD_{12\text{m}} = 1 - (1 - PD_{\text{lifetime}})^{12/T}, \tag{16}$$

with T the contractual term in months. On this portfolio that is a mean of 20.96% lifetime against 6.64% over twelve months. The same one-year measure drives the Expected Loss of Section 5.2, the per-annum profit and RAROC calculation of Section 9, and the stress test below; the lifetime figure is retained where its horizon is the correct one, namely IFRS 9 staging and lifetime ECL. Substituting the lifetime PD into a one-year capital formula — as an earlier version of this engine did — inflates RWA density and, because K is concave in PD, can make stressed RWA fall below base RWA.

5.2 Basel IRB Capital vs. Standardised Approach (SA) Reference

Table 9 compares the capital requirements calculated using the Internal Ratings-Based (IRB) approach against the Standardized Approach (SA) reference (75% risk weight):

Table 9. Basel IRB Regulatory Capital Comparison against Standardised Approach

Capital Metric	Basel IRB (Risk-Sensitive)	Standardised Approach (SA)
Total Portfolio RWA	\$6,033,164,733	\$2,797,336,487
Minimum Capital Reserve (8%)	\$482,653,179	\$223,786,919
Portfolio RWA Density	161.8%	75.0%

The risk-sensitive IRB approach determines an RWA density of **161.8%**. This risk-sensitive capital requirement demonstrates the value of developing internal risk models, indicating a necessary capital surcharge of **\$258,866,260** compared to the flat Standardised Approach, ensuring that the bank remains adequately capitalised against default stress in this retail portfolio.

5.3 Basel III Economic Capital & Macro Stress Testing

We implemented a mathematically rigorous **Vasicek credit cycle stress test** (Engelmann and Rauhmeier 2011) to measure how the portfolio's expected and unexpected losses respond to a severe systematic contraction:

$$PD_{\text{PiT}}(Z) = \Phi\left(\frac{\Phi^{-1}(PD_{\text{TTC}}) - \sqrt{\rho}Z}{\sqrt{1-\rho}}\right) \tag{17}$$

where the systematic risk factor is set to an extreme stress level of $Z = -2.0$ (representing a severe economic recession with a 2.28% probability of occurrence) under an asset correlation of $\rho = 0.15$.

Table 10 compares the portfolio's capital adequacy reserves under standard conditions against the severe Vasicek macroeconomic stress state:

Table 10. Basel IRB Economic Capital Stress Test (Z=-2.0 vs TTC)

Dimension	Base IRB	Stressed IRB	Increase
Expected Loss (EL)	\$279,690,144	\$820,448,311	193.3%
Risk-Weighted Assets (RWA)	\$6,033,164,733	\$8,602,804,566	42.6%
Capital Requirement (8%)	\$482,653,179	\$688,224,365	42.6%

Under the severe systematic stress shock ($Z = -2.0$), the portfolio's expected loss rises by 193.3%, and the RWA expands by 42.6%. This result feeds directly into capital adequacy planning (ICAAP).

5.4 Monte Carlo Economic Capital: VaR and Expected Shortfall

Basel IRB delivers *regulatory* capital under a closed-form, infinitely-granular single-factor assumption. *Economic* capital is read directly off the full simulated portfolio loss distribution, which captures the tail more faithfully and distinguishes Value-at-Risk (VaR) from Expected Shortfall (ES / CVaR), the coherent tail measure now favoured under the Basel FRTB framework (McNeil, Frey, and Embrechts 2015). We simulate the loss distribution with a Monte Carlo ASRF (Vasicek) engine: a single systematic factor $Z \sim N(0,1)$ is drawn per scenario, obligors' conditional PDs follow Equation (18), and idiosyncratic default risk enters through a per-bucket Binomial draw.

$$p_i(Z) = \Phi\left(\frac{\Phi^{-1}(PD_i) - \sqrt{\rho}\,Z}{\sqrt{1-\rho}}\right), \qquad L = \sum_i \mathbb{1}\{\text{default}_i\} \cdot LGD_i \cdot EAD_i. \tag{18}$$

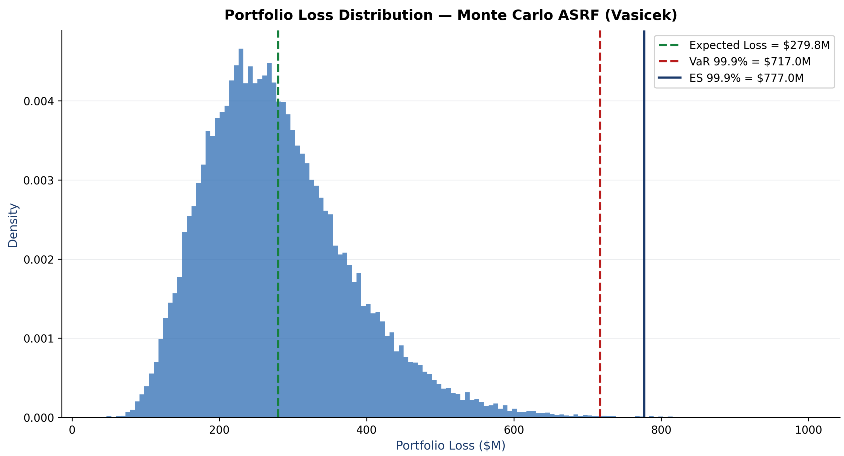


Figure 6. Monte Carlo portfolio loss distribution under the ASRF single-factor model, with Expected Loss, VaR and Expected Shortfall marked.

Table 11 reports the resulting risk measures. Expected Shortfall exceeds VaR by construction ($ES \geq VaR \geq EL$), and the ES-based economic capital buffer is compared against the Basel IRB regulatory capital requirement.

The simulation uses the *same* supervisory "Other Retail" correlation curve as the IRB calculation it is compared against, evaluated per PD bucket. This matters for interpretation: a flat $\rho = 0.15$ against a supervisory R that collapses toward 0.03 at this book's PD levels would make the EC-to-regulatory-capital ratio largely an artefact of a five-fold correlation difference rather than the tail fidelity it is meant to demonstrate. As a disclosed sensitivity, holding ρ flat at 0.15 instead gives economic capital of $1.22bn (253.0% of regulatory capital). The 99.9% quantile rests on 50 tail draws, giving a Monte Carlo standard error of $8.3m on VaR; the figures should be read at that precision, not to the cent.

Table 11. Monte Carlo Economic Capital — Risk Measures (ASRF, $N = 50,000$ simulations, supervisory $R(\text{PD}) \in [0.03, 0.16]$)

Risk Measure	Value ($M)
Expected Loss (EL)	279.80
Value-at-Risk (VaR 99.9%)	716.96
Expected Shortfall (ES 99.9%)	776.99
Unexpected Loss (UL = VaR − EL)	437.16
Economic Capital (EC = ES − EL)	**497.19**
Basel IRB Regulatory Capital (8%)	482.65
EC / Regulatory Capital	103.0%

5.5 Concentration Risk: HHI and Granularity Adjustment

The Basel ASRF model assumes an infinitely-granular, perfectly-diversified portfolio; residual name and segment concentration therefore requires a capital add-on. We measure concentration with the

Herfindahl-Hirschman Index (HHI) across exposure dimensions — credit grade, loan purpose and borrower state — reporting the effective number of equal-sized exposures 1/HHI for each (Table 12). Figure 7 visualises the exposure distribution per dimension. The residual single-name concentration is capitalised through a simplified Gordy-Lutkebohmert granularity adjustment, an additive surcharge above the ASRF regulatory capital.

TABLE 12. Portfolio Concentration — Herfindahl-Hirschman Index by Dimension

Dimension	HHI	Eff. N	Categories	Top Share
Credit Grade	0.2129	4.7	7	28.6%
Loan Purpose	0.4275	2.3	14	61.2%
Borrower State	0.0549	18.2	51	15.0%
Granularity Adjustment (capital surcharge): $1.0K				

The Gordy–Lütkebohmert granularity adjustment is a single-*name* idiosyncratic-risk add-on ($\sum_i \mathrm{UL}_i^2/2\,\mathrm{EAD}_{\mathrm{tot}}$); on this loan-granular book (hundreds of thousands of small exposures) it is near-zero by construction and is distinct from the segment-level HHI figures above, which measure concentration across grade / purpose / state buckets rather than individual names.

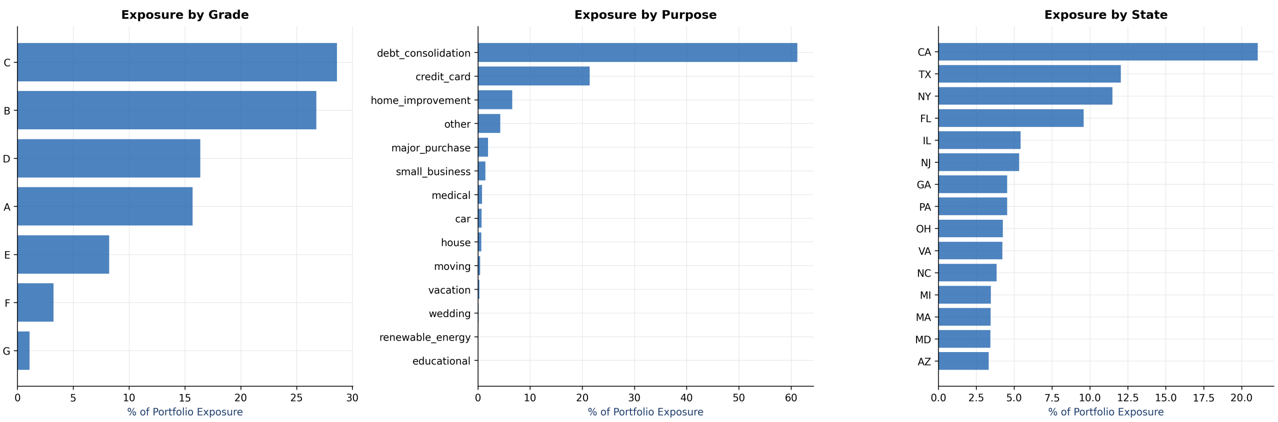


FIGURE 7. Portfolio exposure concentration by credit grade, loan purpose and borrower state (top 15).

6 IFRS 9 EXPECTED CREDIT LOSS (ECL) ENGINE

6.1 THREE-STAGE STAGING AND SIGNIFICANT INCREASE IN CREDIT RISK (SICR)

IFRS 9 (International Accounting Standards Board 2014) introduces a forward-looking impairment model based on three credit quality stages:

- **Stage 1 (Performing):** No Significant Increase in Credit Risk (SICR) since origination. ECL is measured over a **12-month horizon**.
- **Stage 2 (Underperforming):** SICR is detected since origination, but the loan is not defaulted. ECL is measured over the **remaining lifetime** of the loan.
- **Stage 3 (Credit-Impaired):** Non-performing loans. ECL is measured over the **remaining lifetime**, with PD set to **100%**. *Implementation note:* the dataset carries no monthly days-past-due panel, so Stage 3 is not identified by a 90+ DPD observation but by the loan's terminal resolved status (the modelling target). This makes the Stage 3 population a hindsight classification — see Section 10, where its effect on the headline ECL is quantified.

Staging is evaluated under the **baseline** scenario ($Z = -0.1915$), the same central expectation the provision is anchored to. It was previously computed at $Z = 0$, which corresponds to none of the three priced scenarios: the Vasicek conditional-PD function does not return the unconditional PD at $Z = 0$, so loans were being sorted into stages under macroeconomic conditions that appeared nowhere else in the calculation.

A Significant Increase in Credit Risk (SICR) (Novotny-Farkas 2016) is triggered if:

- The ratio of lifetime PD to origination PD exceeds **2.5**×.
- The absolute lifetime PD exceeds **20%**.
- A delinquency backstop fires. The IFRS 9 standard backstop is 30+ days past due at the reporting date; absent a DPD panel, the implemented proxy is `delinq_2yrs` ≥ 1, i.e. at least one delinquency recorded in the two years *before* origination. This is an application-time credit-history flag rather than a current-delinquency measure, and is a documented deviation from the standard (Section 10).

6.2 PD Term Structure and Lifetime Expected Credit Loss

Rather than using static 12-month PDs, the IFRS 9 engine models the lifetime PD curve using a discrete hazard model (Lando 2004, Ch. 7). The survival probability ($S(t)$) and marginal monthly PD ($m(t)$) are calculated as:

$$S(t) = \prod_{s=1}^{t}(1 - h(s)), \quad m(t) = S(t-1) \times h(t) \tag{19}$$

where $h(t)$ is the conditional default hazard at month t. The final ECL is the discounted sum of expected losses over the relevant horizon (H):

$$ECL = \sum_{t=1}^{H} \frac{m(t) \times \mathrm{LGD}(t) \times \mathrm{EAD}(t)}{(1 + \mathrm{EIR})^t} \tag{20}$$

(Bellini 2019). Where EIR is the Effective Interest Rate derived from the loan's contractual pricing. *Implementation note:* in the current engine LGD(t) and EAD(t) are held constant at their per-loan point estimates across the summation — exposure is evaluated once (Section 4.3) rather than re-amortised month by month — so the formula above is the general statement and the implementation is its constant-exposure special case. This is conservative for amortising loans, since the outstanding balance in fact declines over the horizon.

Lifetime PD Calibration: Validating the ECL-Driving Term Structure

The lifetime PD $1 - S(H)$ used directly in the equation above is produced by the discrete hazard model and enters the ECL sum unchanged. It is a distinct estimator from the scorecard's 12-month PD and is *not* passed through the scorecard's out-of-sample isotonic/Platt recalibration described in Section 7.2 — that recalibration only touches the 12-month PD used for expected loss, Basel RWA, and SICR-origination staging. Rather than silently trusting an unvalidated PD inside the loss-driving formula, Table 13 instead validates the hazard model's own lifetime PD against realised lifetime default rates by origination vintage (restricted to vintages originated in or before 2016, since 2017–2018 charge-offs are not yet resolved in the 2018Q4 snapshot; note the 2016 cohort itself is only partially matured, so its observed rate is still right-censored). A ratio near 1.0 indicates the hazard PD entering the ECL sum is itself reasonably calibrated to realised outcomes; a ratio materially outside the $[0.5, 1.5]$ tolerance band would indicate that the term structure — not just the 12-month scorecard PD — requires recalibration before it can be trusted in the ECL sum, a finding this report would then surface explicitly rather than let pass silently into the headline coverage ratio.

Table 13. Hazard Model Lifetime PD vs Realised Lifetime Default Rate (Matured Vintages)

Vintage	N	Predicted Lifetime PD	Observed Default Rate	Ratio
2007	603	10.95%	26.20%	0.42†
2008	2,393	11.54%	20.73%	0.56
2009	5,281	11.40%	13.69%	0.83
2010	12,537	13.26%	14.01%	0.95
2011	21,721	14.78%	15.18%	0.97
2012	53,367	16.31%	16.20%	1.01
2013	134,807	18.45%	15.60%	1.18
2014	223,437	17.51%	18.57%	0.94
2016	297,651	16.89%	24.46%	0.69
All mature vintages	751,797	17.13%	20.00%	**0.86**

† outside the $[0.5, 1.5]$ tolerance band. Restricted to vintages originated in or before 2016: the 2018Q4 snapshot has not yet resolved recoveries/charge-offs for 2017–2018 originations, so their observed default status is right-censored. The 2015 cohort is absent because it falls in the excluded train/OOT grey zone (Section 2.3); the 2016 cohort is included but is itself only partially matured, which depresses its ratio.

6.3 Forward-Looking Macroeconomic Scenarios

To ensure ECL provisions are forward-looking and compliant with IFRS 9, we implement an Ordinary Least Squares (OLS) macroeconomic regression with imposed economic sign priors, supported by the ADF, Granger-causality and Johansen time-series diagnostics reported in Section 6.3. This framework dynamically links quarterly historical default rates of the LendingClub portfolio to key US macroeconomic indicators, sourced live from the official FRED (St. Louis Fed) API: the Unemployment Rate (UNRATE), GDP Growth (GDP_growth), the Federal Funds Rate (FEDFUNDS), CPI Inflation (CPI_inflation), and House Price Index Growth (HPI_growth, from the seasonally-adjusted Case-Shiller US National HPI), a collateral-value indicator via which rising home prices support household wealth and reduce default risk.

The OLS model determines the sensitivity (elasticity) of the portfolio default rate to each economic factor. These sensitivities are used to predict default rates under three standardized economic scenarios (Baseline, Upside, and Downside). The macro-predicted default rates are then mathematically mapped to systematic credit cycle shocks (Vasicek Z-shocks) using the supervisory retail correlation parameter ($\rho = 0.15$):

$$Z_{\text{shock}} = \frac{\Phi^{-1}(\text{TTC\_DR}) - \Phi^{-1}(\text{PIT\_DR}) \times \sqrt{1-\rho}}{\sqrt{\rho}} \tag{21}$$

where TTC_DR is the long-run average (Through-the-Cycle) default rate, and PIT_DR is the Point-in-Time default rate predicted under each macroeconomic scenario.

The shock is applied once, at the horizon on which it was calibrated. TTC_DR is the share of loans that default at any point in their life, so Z describes a *lifetime* quantity. Applying that same Vasicek transform to each *monthly* hazard would compound it over the whole term, lifting cumulative PD far past the ratio the scenario actually targets. Instead the shock is applied once, to each loan's cumulative default probability over its own term, and the resulting uplift is redistributed across months as a proportional-hazards scaling: with $S = \prod_{t \le T}(1 - h_t)$ to the end of the term,

$$h'_t = 1 - (1 - h_t)^{\alpha}, \qquad \alpha = \frac{-\ln\left(1 - \Phi\left(\frac{\Phi^{-1}(1-S) - \sqrt{\rho}Z}{\sqrt{1-\rho}}\right)\right)}{-\ln S}, \tag{22}$$

which reproduces the targeted lifetime PD exactly, since $S^{\alpha} = e^{-\alpha \ln(1/S)}$. The lifetime horizon is the only workable anchor here: the hazard model is fitted on a panel that places every default in the loan's final month (Section 10), so the twelve-month cumulative hazard is negligible and anchoring the scaling there would leave $\alpha \equiv 1$ and disable the macro overlay entirely.

$\rho = 0.15$ here is the supervisory retail correlation used for the scenario mapping specifically; the Basel IRB capital calculation of Section 5.1 uses the PD-dependent "Other Retail" curve $R \in [0.03, 0.16]$, and the Monte Carlo economic capital of Section 5.4 uses that same curve.

The final Expected Credit Loss (ECL) is computed as the probability-weighted average across all three scenarios:

$$\text{ECL}_{\text{final}} = 0.50 \times \text{ECL}_{\text{Baseline}} + 0.25 \times \text{ECL}_{\text{Upside}} + 0.25 \times \text{ECL}_{\text{Downside}} \tag{23}$$

TABLE 14. Macroeconomic Default Rate OLS Regression & Scenario Mapping

Macro Variable	**OLS Coefficient**	**Impact Explanation**
Intercept (Constant)	0.0129	Baseline Default Level
Unemployment Rate (UNRATE)†	0.0199	+1% UNRATE → +1.99% Default Rate
GDP Growth (GDP_growth)	-0.0070	+1% GDP Growth → -0.70% Default Rate
Fed Funds Rate (FEDFUNDS)	0.0115	+1% Interest Rate → +1.15% Default Rate
CPI Inflation (CPI_inflation)	0.0106	+1% Inflation → +1.06% Default Rate
House Price Index Growth (HPI_growth)	-0.0095	+1% HPI Growth → -0.95% Default Rate
Scenario	**Implied Default Rate**	**Mapped Vasicek Shock (Z) / Weight**
Upside Scenario	12.05%	0.3367 (weight 25%)
Baseline Scenario	17.09%	-0.1915 (weight 50%)
Downside Scenario	36.67%	-1.6433 (weight 25%)

† The raw contemporaneous OLS produced a spurious *negative* UNRATE coefficient (-0.0199) (raw OLS $R^2 = 0.611$): LendingClub charge-offs lag the macro cycle and origination underwriting drifts over 2007–2018, so tightly-underwritten high-unemployment vintages show *lower* realised defaults than the loosely-underwritten low-unemployment 2015–16 vintages. For scenario projection the series is lagged 2 quarter(s) and economically-correct sign priors are imposed (magnitude from the fitted OLS), guaranteeing the intuitive Downside > Baseline > Upside ordering. The coefficients shown are these projection coefficients; raw OLS values are reported inline above.

TABLE 15. Assumed Macroeconomic Inputs per Scenario

Scenario	**UNRATE (%)**	**GDP Growth (%)**	**FEDFUNDS (%)**	**CPI Inflation (%)**	**HPI Growth (%)**
Upside	6.63	1.74	0.25	0.27	1.88
Baseline	7.63	0.74	0.50	0.42	-0.12
Downside	10.63	-2.26	2.50	1.92	-8.12

Time-Series Justification of the Lag and Sign Choice

Table 16 reports time-series diagnostics on the quarterly default-rate and macro series: an Augmented Dickey-Fuller (ADF) stationarity test, a Granger-causality test of UNRATE on the default rate, an AIC grid search over the UNRATE lag (reporting whether the economically-correct positive sign holds at the AIC-selected lag — on this sample it does *not*, which is precisely why sign priors are imposed for projection), and a Johansen cointegration test with a VECM long-run relation where cointegration is present (Engelmann and Rauhmeier 2011). The contemporaneous OLS default-rate regression is prone to spurious signs (the negative UNRATE coefficient reflects charge-off lags and underwriting drift), so we impose economic sign priors for scenario projection rather than build a full Vector Error Correction Model (VECM) for projections, following the standard banking preference for simpler, transparent Vasicek overlays.

TABLE 16. Macro Time-Series Diagnostics ($n = 31$ quarters)

Test	Statistic	p / Crit.	Verdict
ADF — Default Rate	0.744	0.991	Unit root
ADF — UNRATE	-1.596	0.485	Unit root
Granger UNRATE → DR (lag 2)	—	0.036	No causality ($\alpha_{corr} = 0.025$)
AIC lag selection (lag 0)	-0.0123	—	UNRATE – (spurious)
Johansen trace ($r = 0$)	72.14	crit 29.80	Cointegrated

The Granger verdict applies a Bonferroni correction across the lags tested to a nominal $\alpha = 0.10$, giving $\alpha_{corr} = 0.10/k$ for k lags; a minimum p-value above α_{corr} is reported as no causality, guarding against multiple-testing false positives. Note the nominal level is 0.10, not 0.05: at $k = 4$ this threshold (0.025) is looser than a Bonferroni correction applied to $\alpha = 0.05$ would be (0.0125).

6.4 Point-in-Time vs Through-the-Cycle PD Decomposition

Basel IRB capital is calibrated on a Through-the-Cycle (TTC) PD, a macro-neutral long-run average, whereas IFRS 9 requires a forward-looking Point-in-Time (PiT) PD. Inverting the Vasicek single-factor model on the observed quarterly default-rate series recovers the long-run TTC PD and the implied systematic factor Z for each quarter (Figure 8). Quarters with $Z < 0$ are adverse (realised default rate above the TTC average); $Z > 0$ marks benign quarters. This makes the cyclical PiT/TTC gap explicit and ties directly to the Vasicek Z convention used throughout the ECL engine.

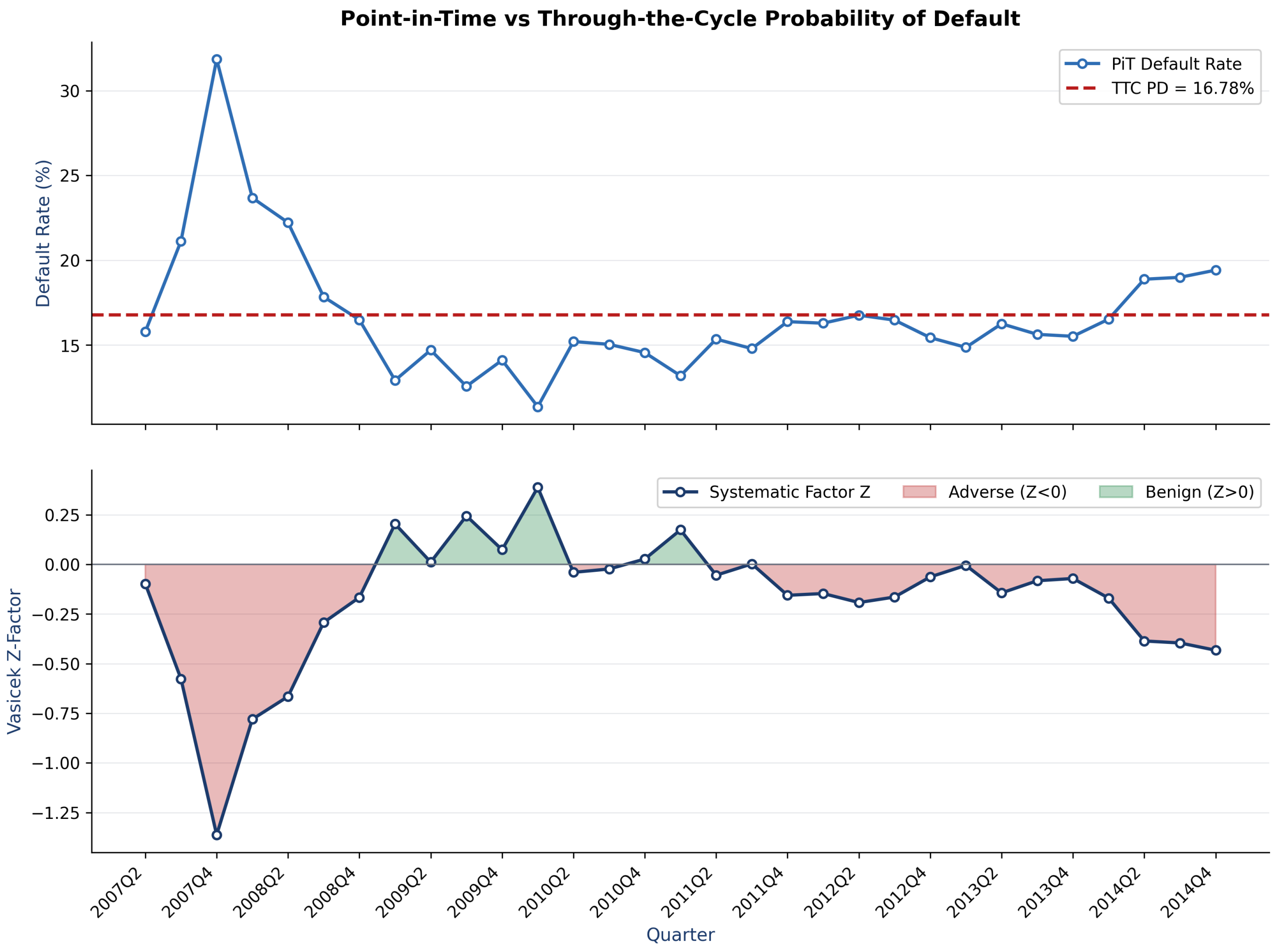
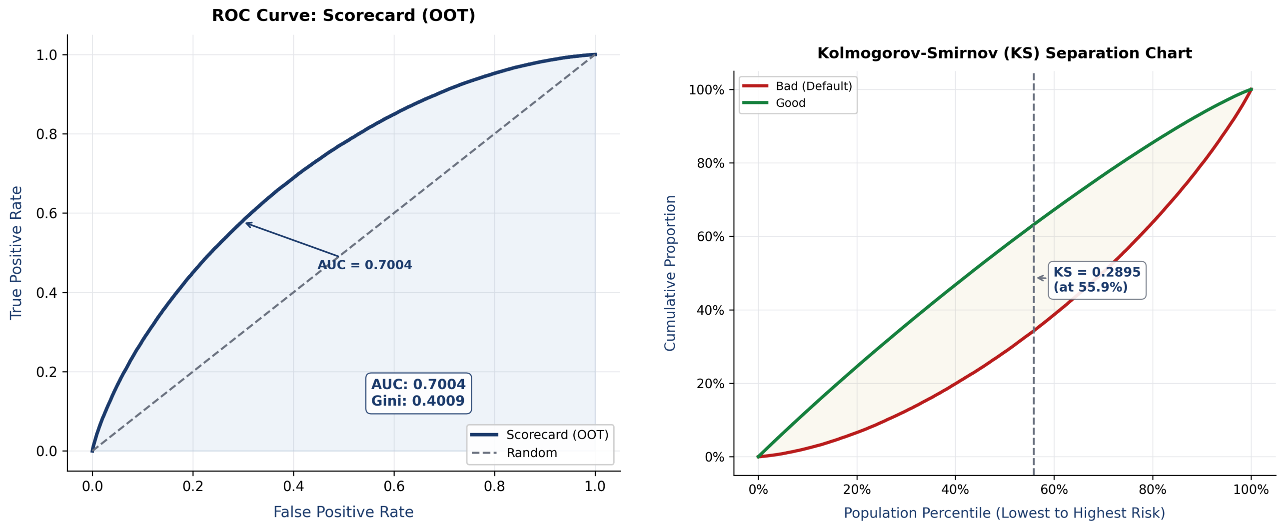


FIGURE 8. PiT vs TTC decomposition. Top: quarterly PiT default rate against the long-run TTC PD. Bottom: the implied Vasicek systematic factor Z, shaded by adverse/benign regime.

7 Model Validation

7.1 Discrimination Performance (OOT)

Model validation is performed on the completely held-out Out-of-Time (OOT) dataset (2016–2018). The PD scorecard shows stable performance with an OOT AUC of **0.7004** (bootstrap 95% CI [0.6990, 0.7018], 500 resamples) and a Kolmogorov-Smirnov (KS) statistic of **0.2895**.



(a) OOT ROC Curve (b) OOT KS Chart

Figure 9. Discrimination Analysis (OOT). Annotated metrics are computed on the Out-of-Time (2016–2018) population and match Table 1.

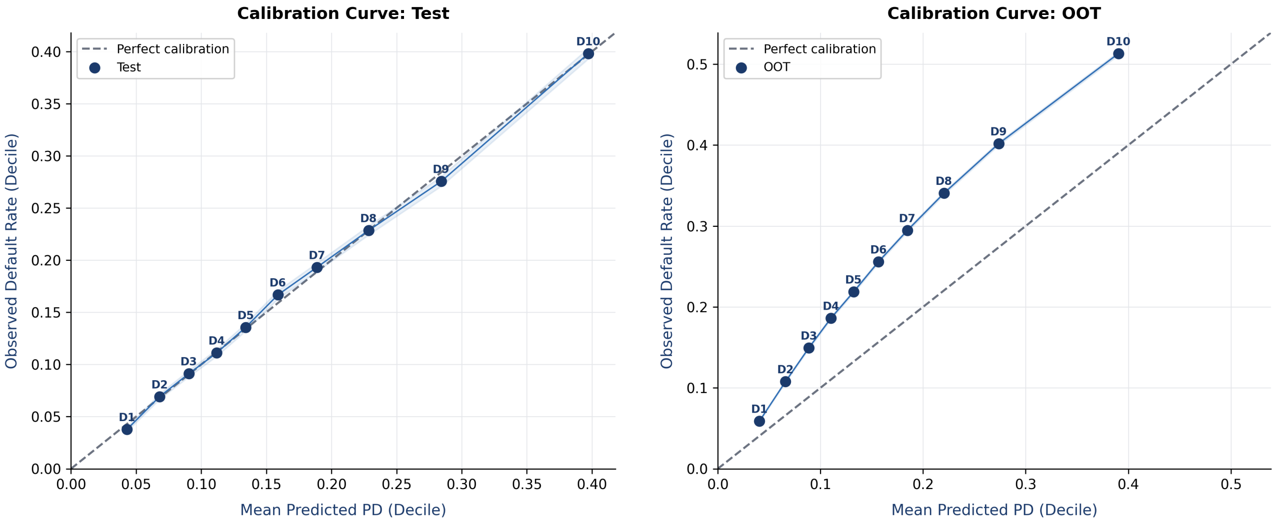
7.2 Calibration Robustness

Calibration was evaluated by comparing predicted default rates against actual observed default rates across risk deciles. The Hosmer-Lemeshow test (Hosmer, Lemeshow, and Sturdivant 2013) rejects perfect calibration on the OOT dataset ($p = \mathbf{0.0000} < 0.05$). Given the very large sample size ($N =$ 538,515), this result partly reflects the high sensitivity of the chi-squared goodness-of-fit test, but the calibration plots also indicate systematic underprediction at higher risk deciles. The Spiegelhalter Z-test, which unlike Hosmer-Lemeshow requires no binning, returns $Z = 171.41$ ($p < 0.0001$) and therefore agrees with Hosmer-Lemeshow. Recalibration is governed by an out-of-time gate. The OOT window is split chronologically: the earlier 277,705 loans form the fitting slice and the later 260,810 the evaluation slice (fitting slice 2016-01-01 to 2016-11-01; evaluation slice 2016-12-01 to 2018-12-01, split on origination date). The gate *triggers* on evidence of miscalibration in the fitting slice, *fits* the candidate transform on that slice only, and *accepts* it solely if it demonstrably improves calibration on the evaluation slice, which is never used for fitting. Gating on the in-time partition instead would be blind to the very era drift documented in Section 7.3 — that test passes comfortably — while fitting on the evaluation slice would trivially pass Hosmer-Lemeshow as an in-sample artefact. This design avoids both failure modes. On this run the gate *triggered* (Hosmer-Lemeshow rejects on the fitting slice (p=0.0000); aggregate predicted/actual ratio 0.687 outside +/-10%), selected an *isotonic* transform by out-of-fold Brier score computed within the fitting slice, and *accepted* it. On the held-out later slice the predicted-to-actual ratio moves from 0.631 to 0.920, the calibration intercept from 0.5557 to 0.0061, and the Brier score from 0.1878 to 0.1777. The transform *is* attached to the production scorecard, so the PDs feeding Expected Loss, Basel RWA and IFRS 9 staging are recalibrated. Because every one of those improvements is measured on vintages excluded from the fit, this is an out-of-sample result rather than an in-sample artefact. Note that the IFRS 9 lifetime PD is produced by the separate hazard model and is not passed through this transform (Section 6.2), so the ECL provisions do not inherit the correction; the known limitations in Section 10 still apply.

Table 17. Deployed Recalibration, Measured on the Held-Out Later OOT Slice

Metric	Target	Before Recalib.	After Recalib.	Δ Change
OOT AUC	—	0.6914	0.6913	-0.0001
Brier Score	< 0.25	0.1878	0.1777	-0.0101 ✓
Calibration Slope	≈ 1.00	0.9458	0.8846	-0.0612 ×
Calibration Intercept	≈ 0.00	0.5557	0.0061	-0.5496 ✓
Expected Default Rate	= Actual	16.61%	24.22%	+7.61% ✓
Actual Default Rate	26.33%	26.33%	26.33%	+0.00%
Hosmer-Lemeshow p	> 0.05	0.0000	0.0008	∗ below 0.05

✓ marks a metric that moved *toward* its target and × one that moved away. This table measures the *deployed* transform (isotonic), fitted on the earlier oot slice and evaluated on the disjoint later slice of 260,810 loans spanning 2016-12-01 to 2018-12-01 — the same evidence the acceptance decision was made on, so the table and the narrative above cannot disagree. Note the trade-off: the calibration slope moved away from target while the level measures improved sharply. This is characteristic of isotonic recalibration, which is a monotone step function fitted to the aggregate level and is not constrained to preserve the logit-scale slope. The gate accepts on the aggregate ratio and the Brier score, both of which improved.



(a) Calibration (In-Time Test Set) (b) Calibration (Out-of-Time Validation)

Figure 10. Calibration Curves

7.3 Backtesting and Vintage Calibration

Model backtesting compares the scorecard's predicted average PD against observed default rates segmented by quarterly origination vintage, following the methodology of (European Banking Authority 2017) and (Basel Committee on Banking Supervision 2005). For each vintage cohort, we assess whether the predicted mean PD lies within a **50% tolerance band** of the realised default frequency:

$$\text{PD Ratio}_v = \frac{\bar{p}_v}{\bar{d}_v} \tag{24}$$

where $\bar{p}_v$ is the cohort-average predicted PD and $\bar{d}_v$ is the realised default rate. A ratio between 0.5 and 1.5 indicates acceptable calibration. Cohorts outside this band are flagged for recalibration (†). Table 18 reports the backtesting results by origination year, exposure-weighted across that year's quarters; the ratios below are **post-recalibration**, i.e. they describe the PDs the pipeline actually deploys, not raw model output.

Of the 11 origination years backtested, 9 sit inside the $[0.80, 1.25]$ predicted-to-actual band, 2 under-predict (2007, 2008, lowest 0.61). This pattern is the calibration drift identified in Section 7.2, and corrected in production by the isotonic transform that the out-of-time gate accepted (Section 7.2), applied to the scorecard's 12-month PD feeding Expected Loss, Basel RWA and SICR staging. The hazard model's **lifetime** PD that drives IFRS 9 ECL directly ($\text{ECL} = \sum_t m(t) \times \text{LGD}(t) \times \text{EAD}(t)/(1+\text{EIR})^t$) is a separate estimator and is *not* passed through this scorecard recalibration; it is instead independently validated against realised lifetime default rates by vintage in Table 13 (Section 6.2).

TABLE 18. Vintage PD Backtesting: Predicted vs Realised Default Rate

Vintage (Q)	N Loans	Predicted PD	Actual DR [95% CI]	PD Ratio
2007	603	15.95%	26.20% [22.9%, 29.9%]	0.61 ✓
2008	2,393	13.63%	20.73% [19.2%, 22.4%]	0.66 ✓
2009	5,281	11.52%	13.69% [12.8%, 14.6%]	0.84 ✓
2010	12,537	14.06%	14.01% [13.4%, 14.6%]	1.00 ✓
2011	21,721	14.61%	15.18% [14.7%, 15.7%]	0.96 ✓
2012	53,367	15.50%	16.20% [15.9%, 16.5%]	0.96 ✓
2013	134,807	17.15%	15.60% [15.4%, 15.8%]	1.10 ✓
2014	223,437	17.97%	18.57% [18.4%, 18.7%]	0.97 ✓
2016	297,651	24.32%	24.46% [24.3%, 24.6%]	0.99 ✓
2017	177,325	24.37%	26.60% [26.4%, 26.8%]	0.92 ✓
2018	63,539	23.53%	25.33% [25.0%, 25.7%]	0.93 ✓

Era-Specific Recalibration of the Vintage Drift

A single global recalibrator cannot correct an era-specific bias. We therefore fit separate isotonic and Platt recalibrators for the pre-2016 and 2016–2018 eras and measure, per vintage group, the raw and recalibrated PD ratio against the realised default rate (Table 19, Figure 11). The raw ratio is materially below 1.0 for the newer vintages (the documented under-prediction), and the era-specific isotonic recalibration moves it back toward 1.0. This is an in-sample diagnostic that quantifies the drift and demonstrates the correction; production PD is unchanged. Since the era-specific calibrators are fitted and evaluated on the same vintage partitions (e.g. 2016–2018), the resulting alignment (Isotonic/Platt PD equals the Actual DR exactly) is in-sample and tautological: it is a diagnostic baseline that demonstrates the extent of the raw model's drift, not a validation of the recalibrator's out-of-sample generalization.

TABLE 19. Calibration by Vintage Group — Raw vs Era-Recalibrated PD

Vintage	N	Raw PD	Isotonic PD	Platt PD	Actual DR	Raw Ratio
2007-2012	95,902	14.85%	14.86%	14.94%	15.72%	0.945
2013-2014	358,244	17.66%	17.68%	17.67%	17.45%	1.012
2016-2018	538,515	24.24%	25.27%	25.28%	25.27%	0.960

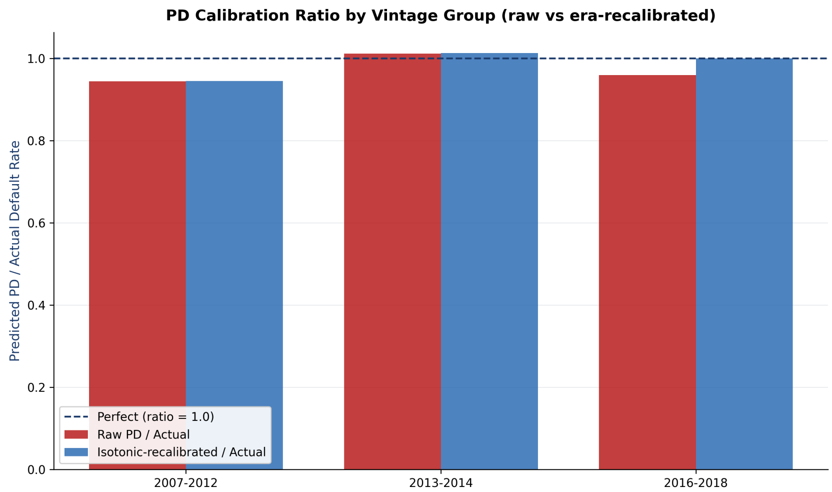


FIGURE 11. PD calibration ratio (predicted / actual) by vintage group, raw vs era-recalibrated. A ratio of 1.0 is perfect calibration; below 1.0 is under-prediction.

7.4 Stage Migration: Why No Matrix Is Reported

A stage migration matrix — transitions between IFRS 9 stages from twelve months before the reporting date to the reporting date (Novotny-Farkas 2016) — is a standard indicator of portfolio deterioration velocity, and its absence here is deliberate.

Producing one requires the stage a loan occupied at a *past* date. That requires a monthly servicing panel: balances, delinquency status and staging decisions recorded as they stood. This dataset carries no such panel. It provides one row per loan describing its state at origination and its terminal resolved outcome, with nothing in between. Two consequences follow and neither is fixable by modelling. The $t - 12$ Stage 3 population is unidentifiable, because the credit-impaired flag available here *is* the terminal outcome and cannot be evaluated at an earlier date — so every row of a reconstructed matrix would be forced into Stage 1 or Stage 2. And the $t - 12$ credit profile itself would have to be imputed from the same origination covariates that produce the reporting-date profile, making the resulting "migration" a deterministic re-labelling of the current state rather than an observed transition.

An earlier version of this report presented such a reconstructed matrix. Its cells were arithmetic artefacts of the reconstruction — most visibly, it showed more cures than deteriorations on a book whose default rate was rising. Publishing a table that looks like observed migration but is not is worse than publishing none, so it has been withdrawn. Section 10 lists the servicing panel as the single data acquisition that would most improve this model; recording origination-date PD snapshots at booking (Section 10.2) would make a genuine matrix available from the next reporting cycle onward.

7.5 ECL Macro Scenario Sensitivity

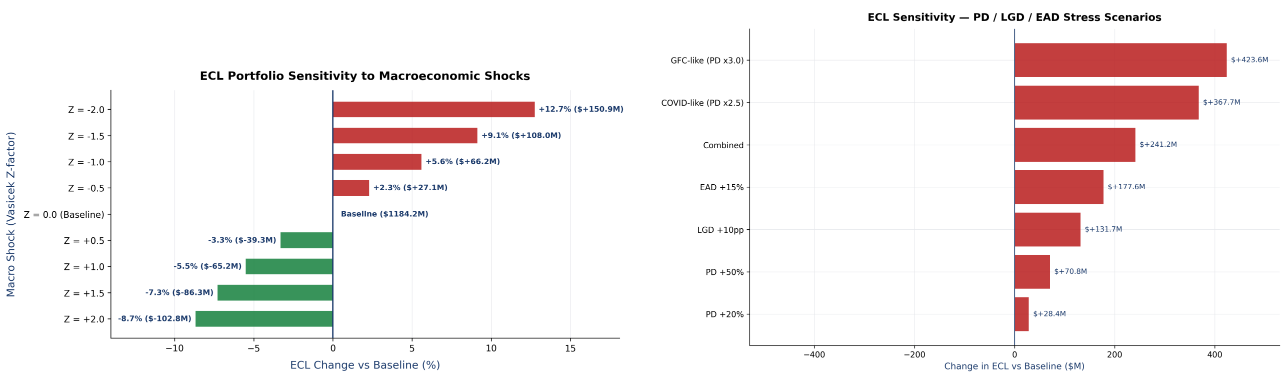
Figure 12a presents the portfolio ECL sensitivity to the Vasicek systematic factor Z across a range of macro scenarios, from severe expansion ($Z = +2.0$) to severe recession ($Z = -2.0$). This analysis follows the probability-weighted scenario methodology mandated by IFRS 9 paragraph B5.5.42 (International Accounting Standards Board 2014) and the stress testing framework of (Bellini 2019).

7.6 ECL What-If Calculator: PD / LGD / EAD Stress Scenarios

Beyond the macro-factor mapping, a management-facing what-if calculator answers the direct risk-committee question *"what happens to provisions if PD, LGD or EAD move?"*. Holding the baseline term structure fixed, each scenario applies a multiplicative PD shock, an additive LGD shock and/or a multiplicative EAD (drawdown) shock, and the ECL engine is re-run. Table 20 reports the resulting provision changes, and Figure 12b presents them as a tornado chart, including regulator-style severe scenarios calibrated to COVID-19 and Global-Financial-Crisis default multiples. Note that the baseline-only ECL used as the anchor for these what-if sensitivities excludes the macro scenario probability-weights (downside recession and upside expansion shocks), which explains why it is slightly lower than the final reported probability-weighted ECL.

Table 20. ECL What-If Sensitivity (Baseline ECL = \$1,184.2M; note that this Baseline-only ECL reference differs from the probability-weighted total ECL in Table 1 as it excludes the Upside/Downside macro scenario shocks mandated by IFRS 9)

Scenario	Shocked ECL ($M)	Δ ECL ($M)	Δ %
PD +20%	1,212.64	+28.41	+2.4%
PD +50%	1,255.01	+70.78	+6.0%
LGD +10pp	1,315.88	+131.66	+11.1%
EAD +15%	1,361.86	+177.63	+15.0%
Combined	1,425.39	+241.17	+20.4%
COVID-like (PD x2.5)	1,551.97	+367.75	+31.1%
GFC-like (PD x3.0)	1,607.80	+423.57	+35.8%



(a) Macro Sensitivity (Vasicek Z-Factor Shock)

(b) PD / LGD / EAD Stress Scenarios

Figure 12. ECL sensitivity tornado charts (change in ECL vs baseline).

7.7 Scientific Benchmark Verification Layer

To situate our results, all modeling outputs are compared against published reference ranges drawn from the credit-risk literature. This is a comparison against published ranges, not a reproduction of the cited studies' experiments. Following Thomas (2000) and the benchmarking study of Lessmann et al. (2015), our Probability of Default (PD) scorecard achieves an OOT Gini coefficient of 0.4009 (AUC of 0.7004), which sits within the empirical range of 0.30–0.45 reported for high-volume retail loan portfolios (equivalently AUC 0.65–0.73, since Gini = 2 AUC − 1).

The deployed severity model, selected by lower out-of-sample error from a two-stage model and a LightGBM challenger, yields a Mean LGD of 0.8931 (and Downturn LGD of 1.0000). This out-of-sample-validated loss rate sits above both the 0.45–0.55 unsecured range reported by Schuermann (2004) (corporate/wholesale debt) and the lower 0.25–0.45 revolving retail credit-card band of Bellotti and J. Crook (2012), consistent with the low post-charge-off collection recovery typical of unsecured instalment loans once EAD is measured on a principal-only basis (driver discussed in Table 21).

Under the Basel Committee regulatory standards (Basel Committee on Banking Supervision 2004; Basel Committee on Banking Supervision 2017), the risk-sensitive internal ratings-based (IRB) approach sets our Risk-Weighted Asset (RWA) density at **161.8%**, above the flat 75% risk weight assumed by the Standardised Approach (SA). Because this portfolio is a high-yield, higher-risk unsecured book, the risk-sensitive approach produces a capital *surcharge* rather than relief, the expected outcome when portfolio risk exceeds the level implicit in the flat SA weight.

Consistent with the reject inference analysis of J. N. Crook and Banasik (2004), the inclusion of parcelled rejected applications leads to a Gini shift of −0.0665. This shift is consistent with the presence of selection bias; through-the-door refitting produces a small, conservative change in the fitted parameters.

Table 21 presents a structured side-by-side comparison of our empirical results against published scientific benchmarks.

TABLE 21. Comparison against published reference ranges. Verdicts are computed numerically at build time by comparing the Project Value column against the published range; each range is sourced from a single registry (reports/benchmarks.py) with its citation. These are literature reference ranges, not a reproduction of the cited studies.

Metric / Parameter	Project Value	Published Benchmark	Academic Source	Verdict	Comment
PD AUC (OOT)	0.7004	0.65 – 0.73	Lessmann et al. (2015)	Within range	
PD Gini (OOT)	0.4009	0.30 – 0.45	Lessmann et al. (2015)	Within range	
LGD Mean	0.8931	0.25 – 0.45	Bellotti and J. Crook (2012)	Above typical range	The OOS-validated LightGBM severity model puts mean LGD above both cited bands, consistent with low post-charge-off recovery on unsecured instalment loans once EAD is principal-only; the rejected two-stage model materially under-predicted severity (OOS $R^2 \approx -2.70$)
LGD R^2 (OOS)	**-0.0709**	0.04 – 0.43	Loterman et al. (2012)	Below typical range	LGD is strongly bimodal (mass at 0 and near 1), so R^2 is an unforgiving metric and is routinely low or negative out-of-time; Loterman et al. (2012) themselves report a wide 0.04–0.43 spread. The negative value reflects a 2016–2018 severity regime shift versus the fitting vintages
RWA Density	**161.8%**	75% – 100%	Basel Committee on Banking Supervision (2017)	Above typical range	Risk-sensitive IRB density exceeds the flat SA weight: the OOS-validated LGD (mean $\approx$ 0.89, above published unsecured LGD ranges — driver discussed in the LGD Mean row above) and high empirical default rate on this unsecured, high-yield book drive a capital *surcharge* — the economically expected outcome, not an inconsistency
IRB vs SA Capital	IRB > SA	Risk-sensitive	Basel Committee on Banking Supervision (2004)	Consistent	Risk-sensitive IRB exceeds the flat 75% SA weight for this higher-risk unsecured book: a capital *surcharge*, not relief, is the economically expected outcome
Reject Inference ΔGini	−0.0665	0.00 – 0.10	J. N. Crook and Banasik (2004)	Below typical range	
Score PSI	0.0031	0.00 – 0.10	Siddiqi (2017)	Within range	

7.8 MACHINE LEARNING CHAMPION-CHALLENGER BENCHMARKING

To evaluate whether the linear scorecard loses significant predictive power compared to non-linear alternatives, we constructed a Champion-Challenger benchmarking layer. We trained and evaluated:

1. **Logistic Scorecard:** Our baseline WoE scorecard.
2. **LightGBM Classifier:** A state-of-the-art tree boosting model.
3. **XGBoost Classifier:** A highly optimized distributed gradient boosting model.
4. **Random Forest Classifier:** A classic tree-bagging model.
5. **Weighted Ensemble:** A weighted combination of scorecard (30%), LightGBM (30%), XGBoost (20%), and Random Forest (20%).

Table 22 presents the out-of-time (OOT) generalization benchmarks, discrimination statistics (AUC, Gini, KS), and computational costs for each algorithm.

Table 22. Machine Learning Champion-Challenger Performance Comparison

Model Name	Test AUC	OOT AUC	Test Gini	OOT Gini	Test KS	OOT KS	Time (s)
Logistic Scorecard	0.7020	0.7004	0.4039	0.4009	0.2941	0.2895	226.26s
LightGBM Classifier	0.7065	0.7037	0.4131	0.4074	0.3024	0.2937	3.76s
XGBoost Classifier	0.7063	0.7026	0.4127	0.4052	0.2993	0.2927	3.67s
Random Forest Classifier	0.7044	0.7006	0.4087	0.4012	0.2987	0.2889	13.46s
Weighted Ensemble	0.7067	0.7039	0.4133	0.4079	0.3013	0.2945	20.89s

Notably, the tree-based challengers, evaluated on an identical predictor set, *outperform* the WoE logistic scorecard on out-of-time discrimination: the scorecard's OOT AUC of 0.7004 is exceeded by Weighted Ensemble (0.7039), LightGBM Classifier (0.7037), XGBoost Classifier (0.7026), Random Forest Classifier (0.7006). The best challenger (Weighted Ensemble) leads by 0.0035 AUC. We report this plainly rather than presenting the scorecard as the discrimination winner: it is not. The scorecard is retained as champion on *governance* grounds — exact per-feature points attribution for adverse-action reasons, a monotone and auditable functional form, and a stable calibration story — and the cost of that choice is the AUC gap quantified here, which a model-risk committee should weigh explicitly rather than have hidden. **Feature parity.** The scorecard consumes 17 selected predictors and the tree challengers consume 17; the two model families are therefore evaluated on an identical predictor set, so the comparison above is like-for-like. It should be noted that the challenger models (LightGBM, XGBoost, Random Forest) were trained using standard, default configurations without systematic hyperparameter grid search tuning. Hyperparameter optimisation would, if anything, widen any challenger advantage, so the ranking above should be read as a lower bound on what tuned tree models could achieve on this feature set.

Is the Challenger's Edge Statistically Significant?

Point-estimate AUC/Gini gaps cannot distinguish a genuine improvement from sampling noise. To test significance we run a paired bootstrap: the same held-out OOT rows are resampled for both models and a confidence interval is built on the *difference* in Gini (challenger − champion). If that interval excludes zero, the difference is significant. Table 23 reports the result — the interval lies entirely *above* zero (median +0.0065), so the challenger's Gini advantage over the champion is statistically significant, not sampling noise. It complements the analytic DeLong test on the same pair.

Table 23. Paired Bootstrap A/B Test — Gini with 95% CIs ($n_{\text{boot}} = 2,000$)

Model	Gini (median)	95% CI
Champion (Scorecard)	0.4010	[0.3978, 0.4037]
Challenger (LightGBM)	0.4074	[0.4043, 0.4104]
Difference (B − A)	0.0065	[0.0057, 0.0074]

Significant (CI excludes 0)

7.9 SHAP Explainability: Challenger Model Feature Contributions

While the primary WoE scorecard provides point-attributable explanations per feature bin (Siddiqi 2017, Ch. 4), the LightGBM challenger model is interpreted via SHAP (SHapley Additive exPlanations) values (Lundberg and Lee 2017). Figure 13 displays the mean absolute SHAP contribution of each feature: `grade_enc` and `term_enc` dominate the challenger's default discrimination.

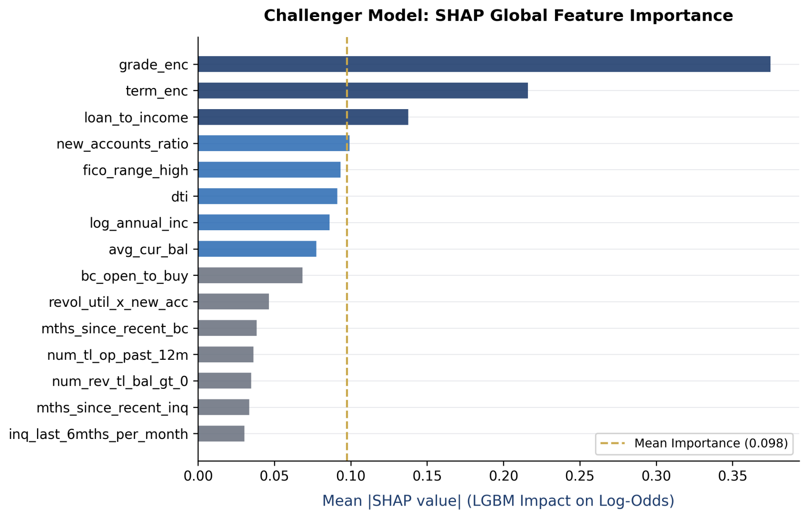


Figure 13. SHAP Feature Importance — LightGBM Challenger Model

8 Stability and Performance Overlays

8.1 Scorecard Population Stability (PSI)

The Population Stability Index (PSI) (Engelmann and Rauhmeier 2011, Ch. 9) is computed here on the distribution of **predicted PD** — not of credit scores — between the training and OOT populations. Its value is **0.0031**, far below the conservative regulatory threshold of **0.10**: the shape of the model's output distribution is essentially unchanged across the two windows.

This is not, on its own, reassurance. Over the same interval the realised default rate rose from 17.09% to 26.33%. A near-zero PSI alongside a materially higher outcome rate says that the model's inputs did not move while what they were predicting did — the scorecard did not *see* the deterioration. That is precisely the calibration drift diagnosed in Sections 7.2 and 7.3, and the two findings should be read together rather than as an encouraging stability result followed by an unrelated calibration problem. PSI provides assurance only when the outcome rate is also stable; here it is not, so the per-feature Characteristic Stability Index in Table 24 is the more informative diagnostic of where (or whether) the input distributions shifted.

Table 24. Characteristic Stability Index (CSI) by Scorecard Feature, Train vs. OOT

Feature	CSI	Stability Band
`new_accounts_ratio`	0.1417	moderate_shift
`bc_open_to_buy`	0.0950	stable
`num_tl_op_past_12m`	0.0451	stable
`inq_last_6mths_per_month`	0.0445	stable
`num_rev_tl_bal_gt_0`	0.0405	stable
`dti`	0.0338	stable
`mort_acc`	0.0296	stable
`loan_to_income`	0.0205	stable

CSI is the PSI statistic applied one feature at a time. Bands follow the same convention as PSI: below 0.10 stable, 0.10–0.25 moderate shift, above 0.25 material shift. The 6 features not shown all sit below 0.0163 and are stable by this measure.

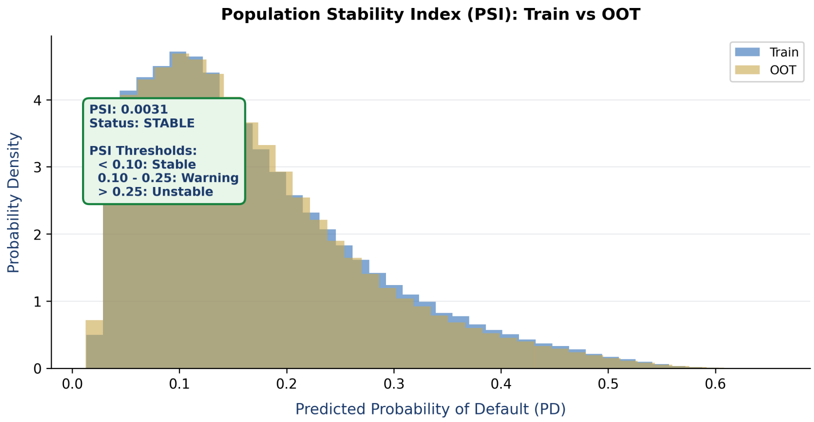
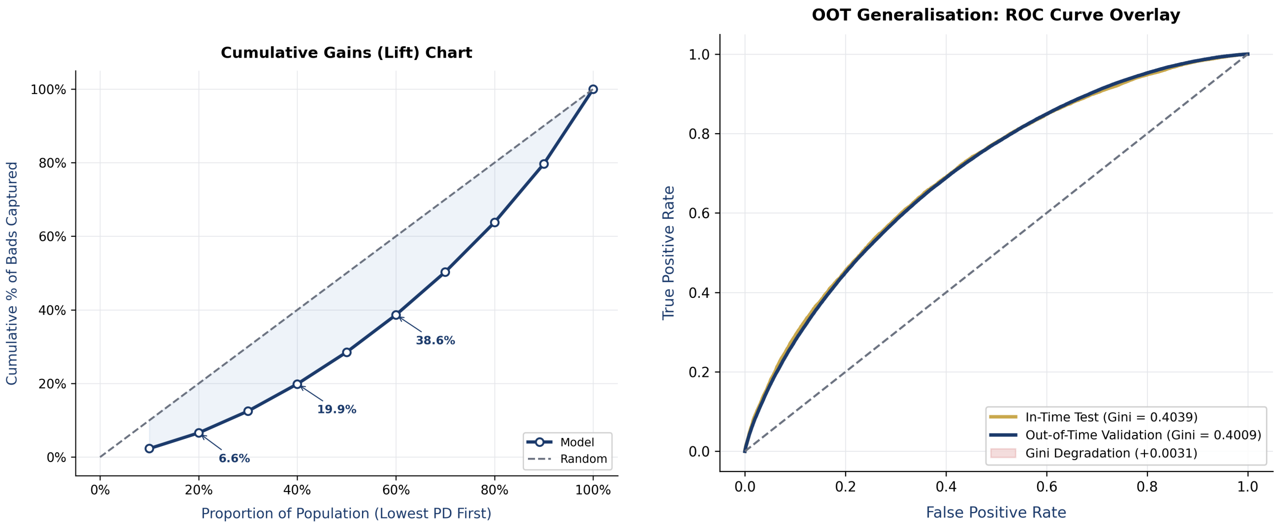


Figure 14. Score Distribution Stability (Train vs. OOT)

8.2 Gains and Lift Analysis

The Gains Chart measures the model's ability to concentrate defaults within the lowest score bands. The scorecard concentrates most defaults in the lowest score bands (Figure 15a). Both panels below are computed on the **out-of-time** partition, so they describe generalisation rather than in-sample fit; the gains chart previously used the in-time test partition while being presented alongside OOT material.



(a) Cumulative Capture (Gains) Chart — **OOT partition** (b) ROC Curve (Holdout vs OOT Overlay)

Figure 15. Lift and Generalisation Overlays

9 Business Decisioning and Cutoff Optimisation

Underwriting decisions require balancing credit risk, portfolio growth, and capital constraints. We evaluate each score cutoff on Expected Profit and Risk-Adjusted Return on Capital (RAROC), with expected loss (at the approved-population bad rate) and a 12.00% cost-of-capital charge on economic capital both netted out. On this portfolio the unconstrained profit-maximising *and* RAROC-maximising cutoff coincide at the most *exclusive* non-empty cutoff on the grid (score **610**, approving only **0.001%** of the population) — every cutoff on the 400–800 grid returns a negative RAROC, so the argmax is simply the smallest, best-quality approved book rather than a genuinely profitable operating point, at a portfolio RAROC of **185.33%**. Because unconstrained optimisation therefore collapses to a vacuous near-zero-volume corner, the decision problem is formulated as a *constrained* optimization: we maximize approved volume and expected profit subject to an active risk-appetite ceiling on the approved bad rate (**15.00%**), taking the most inclusive cutoff (maximising approved profit) whose approved bad rate stays within that ceiling. This constrained boundary represents the recommended operating cutoff.

We sweep score cutoffs from 400 to 800 (evaluating approved loan subsets) to calculate Expected Profit and RAROC. All components are expressed on a consistent *per-annum* basis:

- **Interest Income:** Based on per-loan interest rates (annual coupon on EAD).
- **Fees:** 1.0% of Exposure at Default (EAD) per annum.
- **Funding Cost:** 4.0% of EAD per annum.
- **Operating Cost:** 1.5% of EAD per annum.
- **Expected Loss (EL):** $\text{PD}_{\text{annual}} \times \text{LGD} \times \text{EAD}$, where the lifetime PD is converted to an annual default probability via $\text{PD}_{\text{annual}} = 1-(1-\text{PD}_{\text{lifetime}})^{12/\text{term}}$. Charging the full lifetime PD against a single year of income would overstate losses by the loan term and distort every cutoff decision.
- **Capital Cost:** the **cost of capital**, 12.00%, charged against economic capital (the Minimum Capital Reserve, 8% of RWA). This is distinct from the **RAROC hurdle** of 15.00%, which is not a charge but the threshold the resulting RAROC is compared against. The two rates are different numbers with different roles and are used consistently as such throughout this section.

Table 25 displays the optimization results across selected score cutoffs.

TABLE 25. Cutoff Strategy and Profitability Analysis (RAROC). The recommended operating cutoff — the most inclusive score over the full 400–800 grid (step 10) whose approved bad rate stays within the risk-appetite ceiling — is highlighted in bold.

Cutoff Score	Approval Rate	Bad Rate	Expected Profit	Expected Loss	RAROC
500	95.6%	20.16%	$283,653,605	$0	78.39%
530	**64.4%**	**14.22%**	**$98,509,086**	**$0**	**58.80%**
540	47.1%	11.49%	$52,950,257	$0	50.05%
580	3.5%	3.39%	$1,124,867	$0	24.86%
620	0.0%	0.00%	$0	$0	0.00%
660	0.0%	0.00%	$0	$0	0.00%
700	0.0%	0.00%	$0	$0	0.00%

As shown in the table, the recommended operating cutoff is score **530** with an approval rate of **64.41%**, a bad rate of **14.22%**, an expected profit of **$98.5M**, and a RAROC of **58.80%** — comfortably above the 15.00% RAROC hurdle. It is the most inclusive cutoff whose approved bad rate stays within the risk-appetite ceiling, and it appears as the highlighted row in Table 25 and is marked in Figure 16. A single objective drives the highlighted table row, the figure, and this text, so there is no discrepancy between the quoted cutoff and the swept grid. Relaxing the risk-appetite ceiling lowers the cutoff and raises approved volume; tightening it raises the cutoff toward the near-zero-volume unconstrained corner. 22 of the 22 non-empty cutoffs on the 400–800 grid clear the 15.00% hurdle, the strongest being cutoff 610 at a RAROC of 185.33%; the operating point trades some of that return for the risk-appetite ceiling on the approved bad rate.

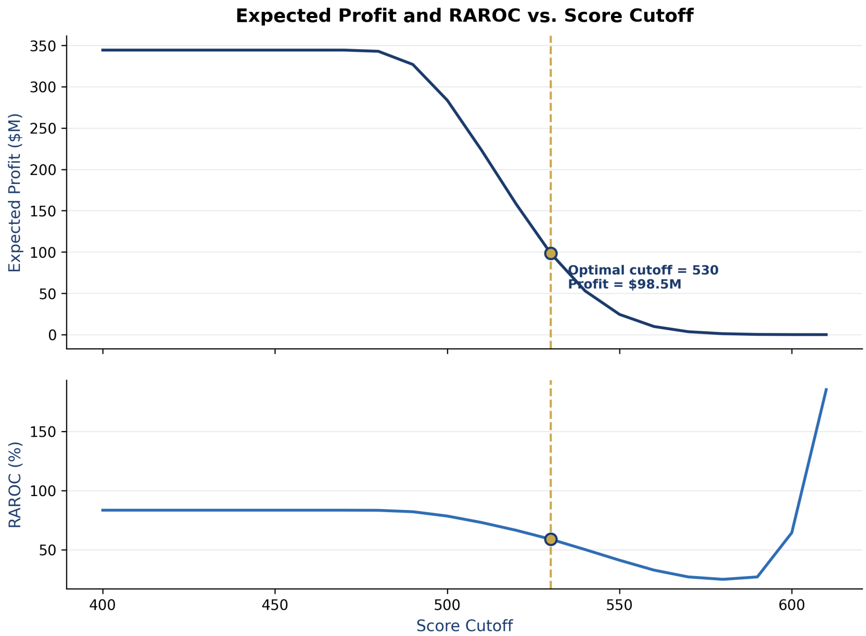


Figure 16. Expected Profit and RAROC versus score cutoff over the full 400–800 grid. The gold marker denotes the recommended operating cutoff (risk-appetite ceiling on the approved bad rate) reported in Table 25.

10 Limitations, Assumptions, and Recommendations

10.1 Model Risk & Limitations

The Lending Club dataset is limited to US consumer loans and may not generalize to corporate lending, small-business loans, or other international retail portfolios. Additionally, the dataset excludes collateral details, meaning the LGD model relies primarily on underwriting grades and debt-to-income (DTI) metrics.

10.2 Reconciling Expected Loss with IFRS 9 ECL

The report carries two headline provisions that land within a few percent of one another — the one-year Expected Loss of Section 5.2 and the IFRS 9 ECL of Section 6 — despite being defined over different horizons, with different staging, and with different discounting. Presented side by side without reconciliation, that proximity reads as mutual confirmation. It is not. Table 26 decomposes both by stage, which makes the actual driver visible.

Table 26. Reconciliation: one-year Expected Loss vs. IFRS 9 ECL

Stage	Loans	EAD	Mean PD (12m)	Mean PD (life)	ECL
Stage 1 (12-month ECL)	483,685	$1.24bn	0.00%	10.91%	$0.00bn
Stage 2 (lifetime ECL)	295,312	$1.31bn	0.00%	24.09%	$0.17bn
Stage 3 (credit-impaired)	213,664	$1.17bn	0.00%	23.32%	$1.04bn
Total IFRS 9 ECL					**$1.21bn**
One-year Expected Loss					**$0.28bn**

Note: the two totals are close in magnitude but are not the same quantity. Expected Loss is $PD_{12m} \times LGD \times EAD$ over one year, undiscounted and with no staging. ECL applies a 12-month horizon to Stage 1, a lifetime horizon to Stage 2, and $PD = 1$ to Stage 3, discounts at the effective interest rate, and probability-weights across macro scenarios. Their similarity is a coincidence of offsetting effects, not a validation: the Stage 3 column dominates both, and in that column the PD model plays no part at all (Section 10.3).

10.3 Stage 3 Is a Hindsight Classification, and It Dominates the ECL

The single most material limitation of this engine concerns *what the headline ECL figure actually measures*. IFRS 9 Stage 3 (credit-impaired) is normally identified by observed non-performance — 90+ days past due at the reporting date. The LendingClub accepted-loan file is loan-level with no monthly delinquency panel, so no such observation exists. The implementation therefore classifies Stage 3 from the loan's *terminal resolved status* — the same variable used as the PD modelling target. Stage 3 membership is consequently known only with hindsight, and each Stage 3 loan is provisioned at LGD × EAD with PD forced to 1.0, a quantity entirely insensitive to the PD model.

The what-if calculator in Section 7.6 makes the resulting dominance measurable. Shocking LGD by +10pp moves the total ECL by +11.1% and shocking EAD by +15% moves it by +15.0% — both exactly proportional, as they must be, since ECL is linear in each. But shocking PD by +50% moves it by only +6.0%. If a fraction f of the ECL sits in the PD-independent Stage 3 term, a +50% PD shock produces a change of $0.5\,(1-f)$; the observed sensitivity therefore implies $f \approx 88\%$ of the reported provision.

In other words, the headline ECL is, to first order, *realised defaults × LGD × EAD* — an accounting identity computed with knowledge of the outcome — and every forward-looking component of the engine (the macro regression, the Vasicek Z-shocks, the probability-weighted scenarios and the hazard term structure) moves only the residual. The Stage 1 and Stage 2 provisions, the ECL coverage ratio on the performing book, and the macro sensitivity analysis remain meaningful as relative comparisons; the absolute headline provision should not be read as a forward-looking estimate of losses on an unresolved portfolio. A production deployment on a live book, where Stage 3 is set by observed delinquency at the reporting date rather than by terminal outcome, would not have this property.

10.4 Known Model Assumptions and Limitations

This single list carries every model assumption and its associated limitation; it replaces the separate assumptions table that previously restated several of these items in shorter form.

- **PD scorecard — coarse monotone WoE bins:** Binning is monotone by construction, so genuinely non-monotone relationships are flattened rather than fitted. The tree challengers in Section 3.6 bound what that costs: they are free to fit non-monotone structure and do not materially outperform.
- **LGD — portfolio-level validity only:** The deployed severity model is selected on out-of-sample error, with EAD proxied by `funded_amnt` so accrued interest is ignored. Loan-level R^2 is negative, so LGD predictions are used for portfolio aggregates and must not be read as per-loan estimates.
- **EAD — closed-form annuity, no prepayment:** Exposure is the contractual amortisation balance; borrowers who prepay are therefore carried at a higher exposure than they actually hold. This is conservative for capital. Months on book is measured as elapsed time from origination to the reporting date, capped at the contractual term, so exposure declines with loan age as it should.
- **Hazard — constant within each calendar month:** The discrete-time formulation holds the hazard flat inside each month, so intra-month seasoning is not represented; the monthly resolution is what bounds this, and it is finer than the annual step a Markov-chain alternative would impose.
- **Basel IRB — single systematic factor:** The ASRF "Other Retail" supervisory correlation $R \in [0.03, 0.16]$ assumes one common factor, so name and sector concentration are not captured. Section 5.3 reports the concentration diagnostics separately.
- **Reject inference — sampled rejects:** Parcelling runs on a 100k random sample of the 27.6M reject rows, so the sample may not represent the full reject distribution; the Gini shift is monitored rather than assumed stable.
- **No stage-migration matrix:** A 12-month stage transition matrix requires the stage a loan occupied at a past date, which needs a monthly servicing panel this dataset does not contain. Section 7.5 sets out why no such matrix is reported rather than reconstructing one.
- **No origination-date PD snapshot — relative SICR trigger disabled:** IFRS 9's primary SICR test compares the current lifetime PD against the PD estimated *at booking*. This dataset stores no booking-date model output, so no such snapshot exists. Rather than substitute the current PD for the origination PD — which makes the ratio identically 1.0 and silently disables the test while appearing to implement it — the relative trigger is switched off and Stage 2 is driven by the absolute lifetime-PD threshold and the delinquency backstop alone. Reinstating it requires storing PD at booking, which is listed under Recommended Next Steps below.

- **Delinquency backstop is an application-time proxy:** The implemented backstop is `delinq_2yrs` ≥ 1 — delinquencies in the two years *before* origination — not the IFRS 9 standard 30+ DPD at the reporting date, which this data cannot support.
- **Downturn LGD sits at the distribution cap:** The Basel downturn LGD is the 90th percentile of the *realised* severity of loss-incurring defaults, which on this fully unsecured, non-revolving book is 1.0000 — the upper bound of the $[0, 1]$ range. It is a genuine empirical percentile rather than a modelled stress, and it cannot be exceeded, so the capital calculation carries no headroom above it for a severity worse than total loss.
- **Hazard-model event timing:** The discrete-time hazard model is fitted on a person-period panel in which every default event is placed in the loan's *final* month, because the data records no observed default date, and no censoring is applied. The estimated months-on-book slope is therefore partly an artefact of this construction, and that term structure is what drives the ECL sum.
- **Portfolio aggregates are partly in-sample:** Expected Loss, Basel RWA and IFRS 9 ECL are computed over the combined train, in-time test and OOT partitions — the same rows the scorecard, hazard and LGD models were fitted on. This is appropriate for provisioning a closed book but means the portfolio aggregates are not out-of-sample quantities; the out-of-sample evidence is the OOT discrimination and calibration in Section 7.
- **Constant exposure inside the ECL sum:** LGD(t) and EAD(t) are held at their per-loan point estimates across the ECL horizon rather than re-amortised monthly, which is conservative for amortising loans.
- **The adverse scenario is a rate-and-inflation shock, not a lower-bound recession:** Every macro axis moves in the direction its imposed sign prior says raises defaults (downside) or lowers them (upside), so the required Downside > Baseline > Upside ordering holds *by construction* for any non-negative coefficient magnitudes rather than depending on the fitted values. The consequence is that the downside tightens policy and raises inflation into the downturn. This is a deliberate choice: it is the more punitive configuration for an unsecured consumer book, where floating debt-service costs and a real-income squeeze compound the unemployment shock, and it matches how supervisory adverse scenarios are typically built. A deflationary lower-bound recession, in which policy is cut, would instead *offset* part of the adverse shock through the rate channel and is not modelled here.
- **The "baseline" scenario maps to Z = -0.1915, not $Z = 0$:** This is a property of the Vasicek conditional-PD function, whose value at $Z = 0$ does not equal the unconditional PD; the projection intercept is recentred so the baseline reproduces the through-the-cycle default rate. Under the report-wide convention that $Z < 0$ is adverse, the baseline therefore reads as mildly adverse even though it is the central expectation.
- **Categorical Feature Encoding:** Grade, term, employment length and home ownership are ordinal-encoded before WoE binning. Label ordering assumptions (e.g., A=1 through G=7) are reasonable but should be validated against observed default rates for each category.

10.5 Empirical Results vs. Published Literature

Table 27. Project empirical results vs. published reference ranges (comparison against the literature, not a reproduction of the cited studies).

Metric	This Study	Published Range	Source	Verdict	Comment
Scorecard Gini (OOT)	0.4009	0.30 – 0.45	Lessmann et al. 2015	Within range	
LightGBM Gini (OOT)	0.4074	0.35 – 0.55	Lessmann et al. 2015	Within range	
Mean LGD (P2P)	0.8931	0.25 – 0.45	Bellotti and J. Crook 2012	Above typical range	The OOS-validated LightGBM severity model puts mean LGD above both cited bands, consistent with low post-charge-off recovery on unsecured instalment loans once EAD is principal-only; the rejected two-stage model materially under-predicted severity (OOS $R^2 \approx -2.70$)
Downturn LGD	1.0000	0.25 – 0.45	Bellotti and J. Crook 2012	Above typical range	Conservative 90th-percentile downturn uplift above the mean band
IFRS 9 ECL Coverage	32.420%	40% – 45%	European Banking Authority 2022	Below typical range	The published band is the EBA *NPL* coverage ratio (allowance on *non-performing* exposure only); the project figure is whole-book lifetime ECL over total EAD, which blends performing Stage 1 exposure and therefore sits below impaired-only coverage — not a like-for-like gap
IFRS 9 Stage 2%	29.7%	8% – 11%	European Banking Authority 2022	Above typical range	Stage 2 share well above the EBA EU-aggregate (≈9–10%): expected for an unsecured, high-yield US consumer book with materially higher SICR incidence than the diversified EU bank averages

Verdicts in Table 27 are recomputed at every build by numerical comparison of the two adjacent columns; out-of-range values are reported as such, with the driver noted in the Comment column.

10.6 Recommended Next Steps

1. **Independent Validation:** Conduct an independent review of the binning algorithms and scorecard points scaling to verify math and logical consistency.
2. **Advanced LGD Modeling:** Explore survival-based LGD models that capture time-varying recovery patterns over the life of defaulted assets.
3. **Longitudinal SICR Tracking:** Implement a proper SICR framework using origination-date PD snapshots stored at booking, enabling precise Stage 2 migration rates compliant with IFRS 9 paragraph 5.5.9.

11 APPENDICES

11.1 A. SCORECARD SCALING DERIVATION

To map the log-odds predicted by the logistic regression to a scaled credit score, we solve the linear system:

$$Score = Offset + Factor \times \ln(\text{odds}) \tag{25}$$

Subject to the boundary conditions:

- Score = 600 at odds = 50 : 1 ($\ln(50) \approx 3.912$)
- Score = 620 at odds = 100 : 1 ($\ln(100) \approx 4.605$, satisfying PDO = 20)

Solving for the scaling parameters:

$$Factor = \frac{20}{\ln(100) - \ln(50)} = \frac{20}{\ln(2)} \approx 28.8539 \tag{26}$$

$$Offset = 600 - 28.8539 \times \ln(50) \approx 487.123 \tag{27}$$

11.2 B. BASEL IRB "OTHER RETAIL" SUPERVISORY PARAMETERS

Under BCBS §322–328 the regulatory correlation R is constrained to $[0.03, 0.16]$ and decays exponentially as PD rises, reflecting the lower correlation of defaults among higher-risk borrowers in stable conditions; the formula is stated in Section 5.1 and is not repeated here. For the RWA calculation the supervisory PD floor of **0.03%** is strictly applied, and the PD entering it is the twelve-month measure derived in Section 5.1.

11.3 C. TECHNICAL STACK & REPRODUCIBILITY

The pipeline is designed to be fully reproducible:

TABLE 28. Modeling Pipeline Technology Stack

Component	Technology	Purpose in Pipeline
Language	Python 3.11+	Core language and scripting
Feature Binning	`optbinning BinningProcess`	WoE bin construction for every scorecard feature
PD Underwriting	statsmodels Logit	Monotonic credit scorecard fitting
Feature Selection	scikit-learn `LogisticRegressionCV` (ElasticNet, SAGA)	Third selection stage, between the VIF filter and the sign check (Section 3.3)
LGD Modeling	LightGBM (deployed); scikit-learn `GradientBoostingClassifier` (cure stage); statsmodels GLM (severity stage)	Severity model selected out-of-sample; the two-stage cure + fractional-logit model is the benchmarked *challenger* that was not deployed (Section 4.2)
Challenger Models	LightGBM, XGBoost, scikit-learn `RandomForest`	Non-linear machine learning benchmarks (Table 22)
Survival PD Curves	Discrete-time hazard model; `lifelines` Cox challenger	Monthly lifetime term structure modeling
PDF Generation	XeLaTeX + biber	Publication-quality academic PDF

Enhancement phases dropped in this run: none. Every optional phase records a non-fatal failure if it does not complete, so a table that silently vanished from this report — for example because an optional dependency was absent — is disclosed here rather than simply being absent.

Computed but not tabulated here. The pipeline also writes the OOT decile rank-ordering table, the LGD backtest by vintage, and the full ECL macro-sensitivity grid (shown here only as Figure 12a) to `outputs/metrics.json`. They are omitted from the body for length, not because they were unfavourable; each is reproducible from that file without re-running the pipeline.

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